Advancing GeoAI for Disaster Resilience:
From Damage Assessment to Insightful Decision Support

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Committee members:

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11.13.2026 @ Proposal Defense, Online

"Seeing, describing, and governing disasters — beyond the boundaries of the screen, let the real world become the stage for intelligence." --Yifan Yang

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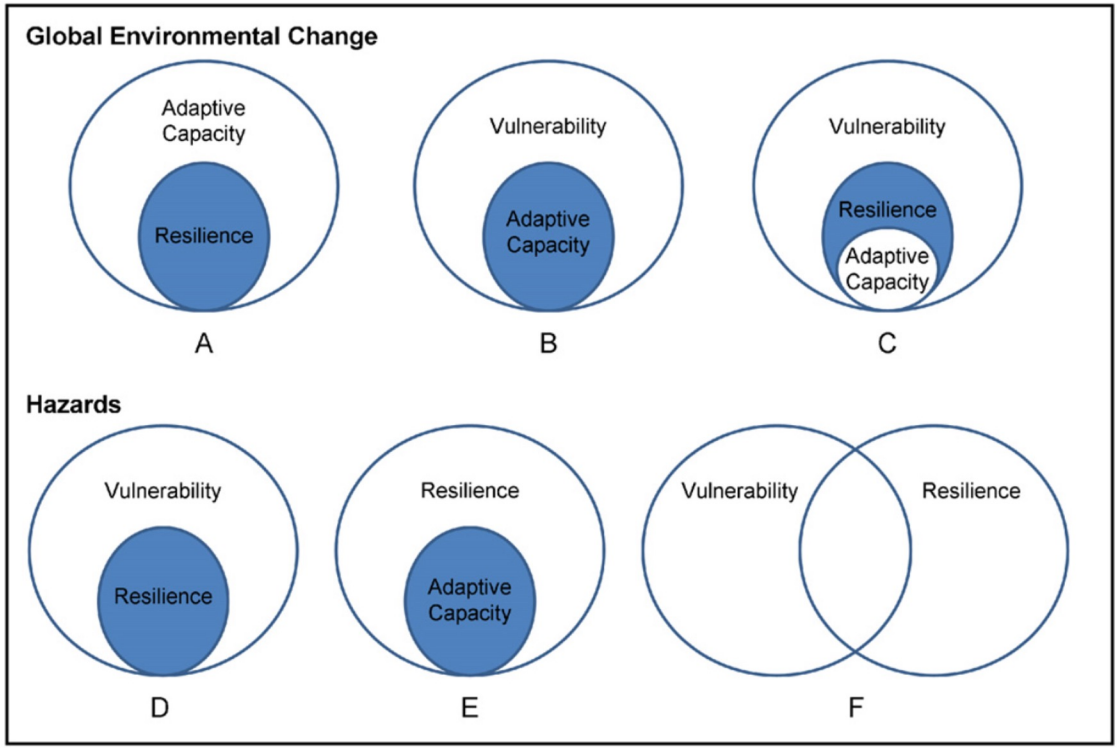
Outline

- Background
- Current Challenges
- Conceptual Framework
- Case Studies (Stages 1–3)
- Publications and Awards
- Research Schedule

Background

Disaster resilience is a geographical question

(1) Rooted in space: resilience is place-based, shaped by local hazard exposure, social vulnerability, and human–environment interactions (Cutter et al., 2008).

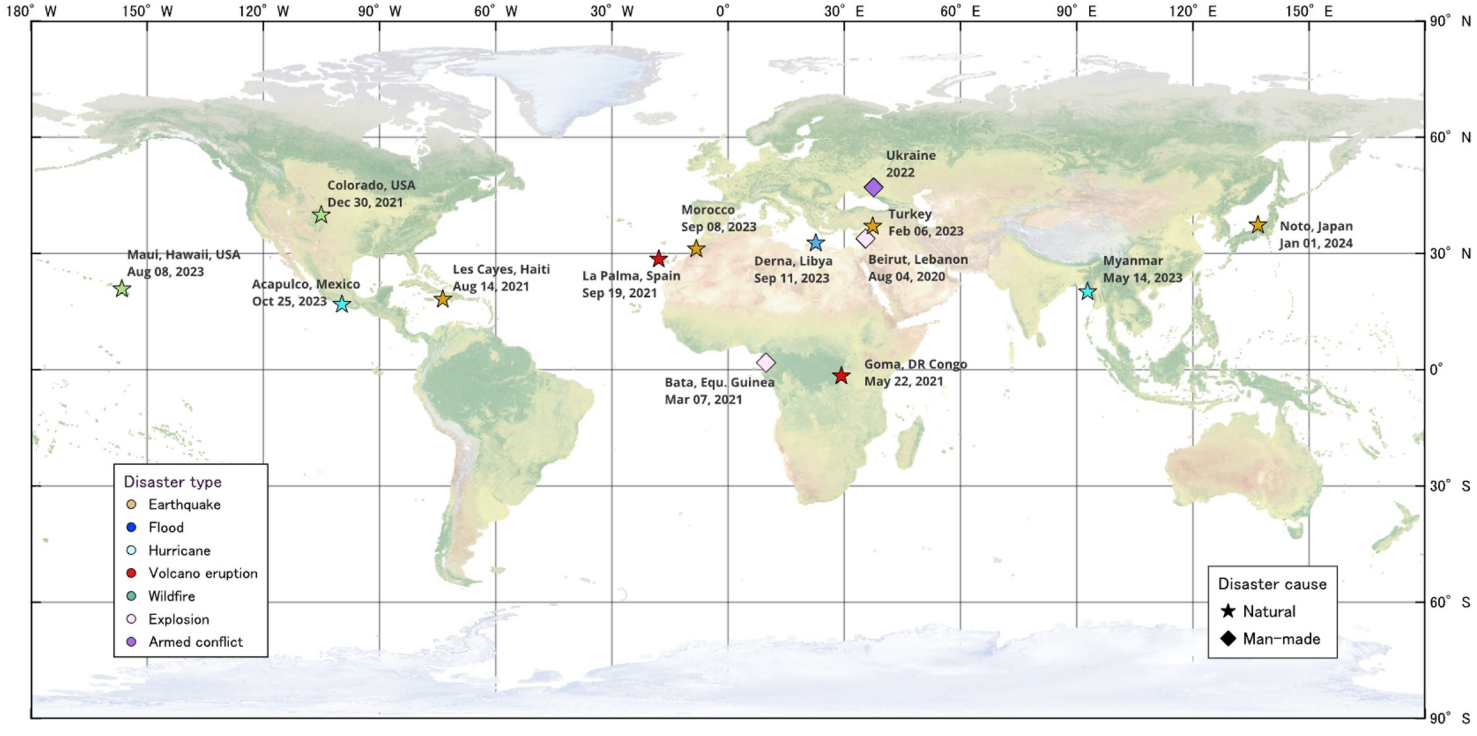


Cutter, S. L., Barnes, L., Berry, M., Burton, C., Evans, E., Tate, E., & Webb, J. (2008). A place-based model for understanding community resilience to natural disasters. Global Environmental Change, 18(4), 598–606.

Background

Disaster resilience is a geographical question

(2) Diverse across the globe: disasters strike different places with different hazards and impacts. The 14 BRIGHT events alone span five continents and seven disaster types, both natural and man-made (Chen et al., 2025).

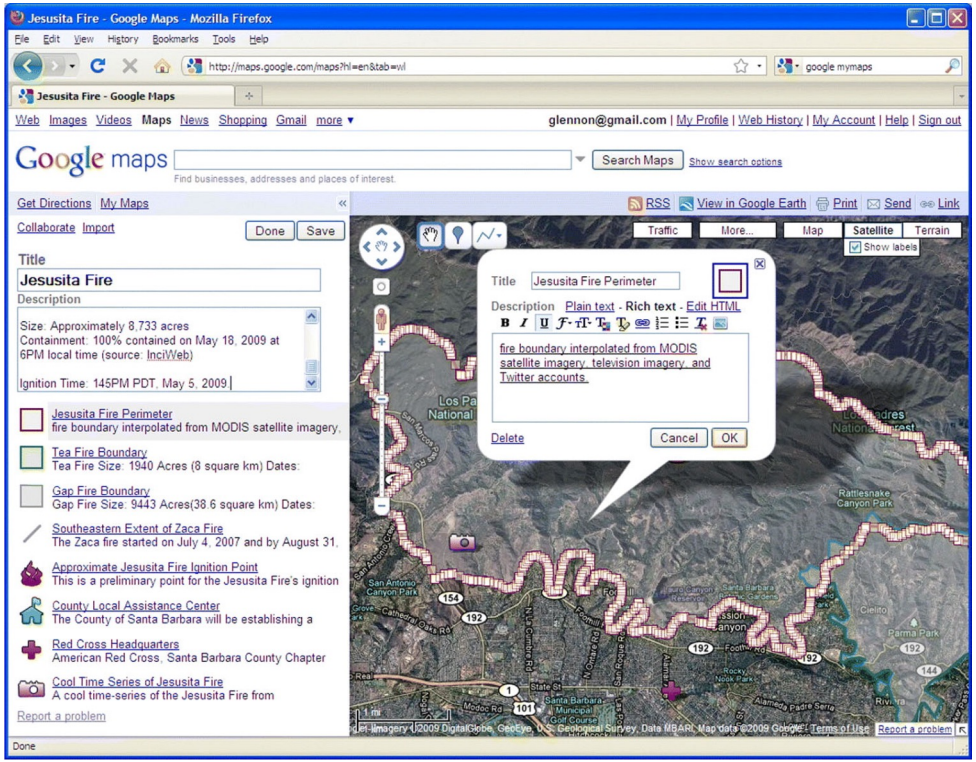


Chen, H., Song, J., Dietrich, O., Broni-Bediako, C., Xuan, W., Wang, J., ... & Yokoya, N. (2025). BRIGHT: A globally distributed multimodal building damage assessment dataset with very-high-resolution for all-weather disaster response. Earth System Science Data, 17(11), 6217–6253.

Background

Disaster resilience is a geographical question

(3) A geographic inquiry: beyond what happened, we ask where, how severe, who was affected, and how to allocate resources (Goodchild & Glennon, 2010).



Goodchild, M. F., & Glennon, J. A. (2010). Crowdsourcing geographic information for disaster response: a research frontier. International Journal of Digital Earth, 3(3), 231–241.

Background

Disaster resilience is a geographical question

(4) A cycle in time: resilience is the ability to prepare for, absorb, recover from, and adapt to adverse events (National Academies et al., 2012). Each phase, from preparedness to response, recovery, and mitigation, depends on timely, place-specific information.

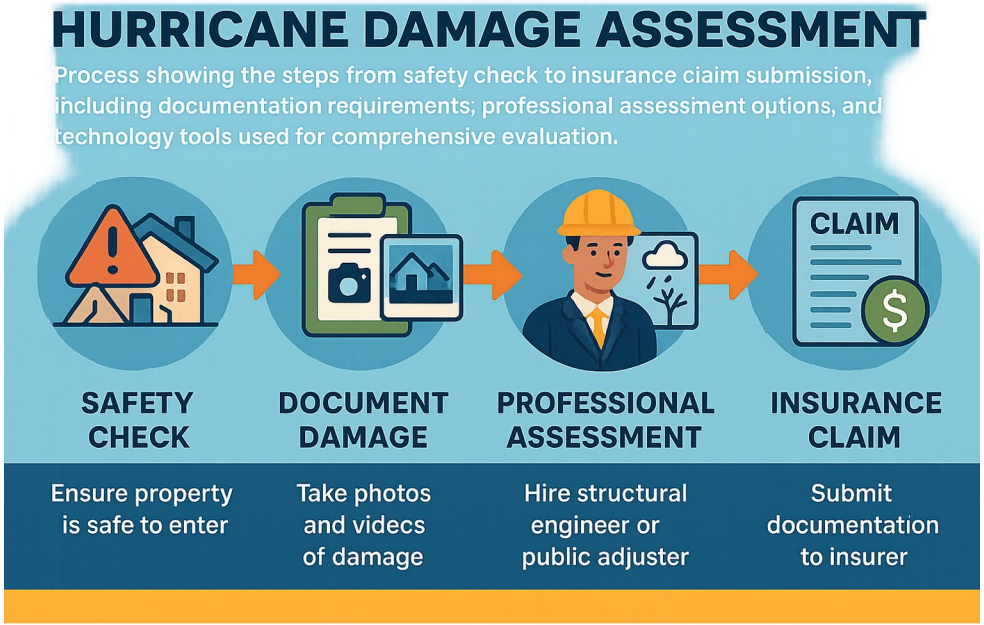
Image generated by ChatGPT

National Academies, Policy, Global Affairs, Committee on Science, Public Policy, & Committee on Increasing National Resilience to Hazards. (2012). Disaster resilience: A national imperative. National Academies Press.

Background

Disaster assessment is a critical task for building resilience

Core tasks: acquire imagery (remote sensing + street view) → classify severity → generate reports → map impacts rapidly (FEMA, 2016).



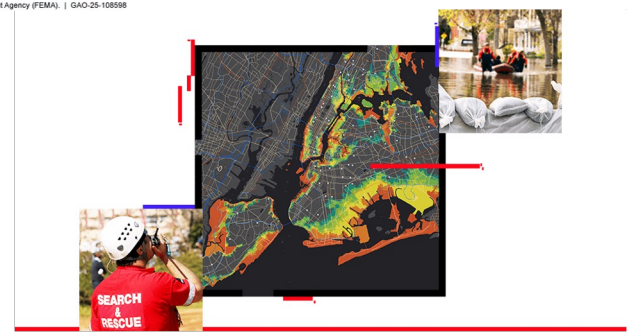
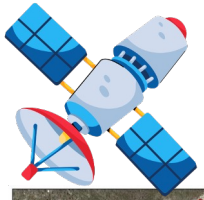


Image source: Insurance Claim Recovery Support, hurricane damage assessment workflow. · FEMA (2016). Damage Assessment Operations Manual: A Guide to Assessing Damage and Impact. U.S. Department of Homeland Security.

Background

Image data is central to disaster assessment: two complementary views

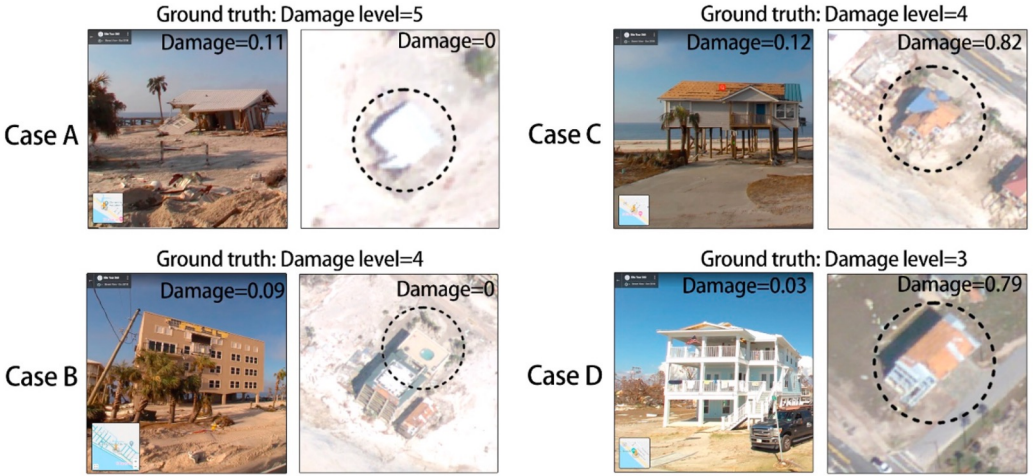


Overhead view — remote sensing

Satellite and aerial imagery: wide coverage and roof-level damage (e.g., xBD benchmark; Gupta et al., 2019).

Ground-level view — street-view imagery

Facade-level detail of structures, debris, and flooding, captured by survey vehicles, panoramic cameras, and volunteers (Zhai & Peng, 2020).



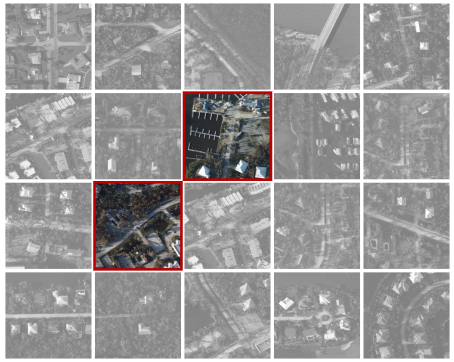
Left: Hauptman, L., Mitsova, D., & Briggs, T. R. (2024). Hurricane Ian damage assessment using aerial imagery and LiDAR: A case study of Estero Island, Florida. Journal of Marine Science and Engineering, 12(4), 668.

Right: Zhai, W., & Peng, Z. (2020). Damage assessment using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. Applied Geography, 123, 102252.

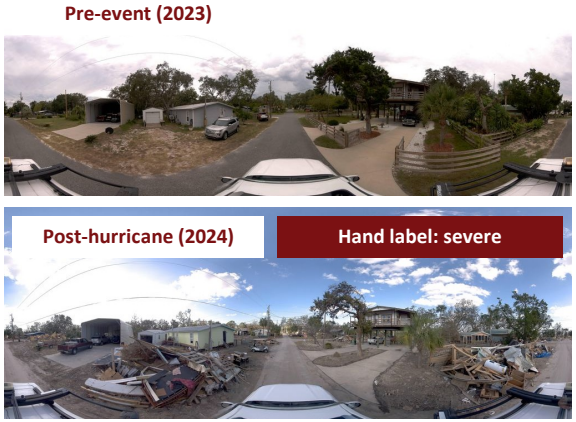
Gupta, R., Hosfelt, R., Sajeev, S., Patel, N., Goodman, B., Doshi, J., Heim, E., Choset, H., & Gaston, M. (2019). xBD: A dataset for assessing building damage from satellite imagery. CVPR Workshops.

Background

The bottleneck: imagery is abundant, interpretation is not



Captured, never analyzed **Actually analyzed**



Volume (too much)

Satellites capture every tile of a disaster zone within days, but only a few tiles are ever analyzed

Speed (too slow)

Labels are made by hand, one image pair at a time: 2,556 pairs for one small town

Complexity (too difficult)

The same wildfire-burned building, seen from two views, tells two stories

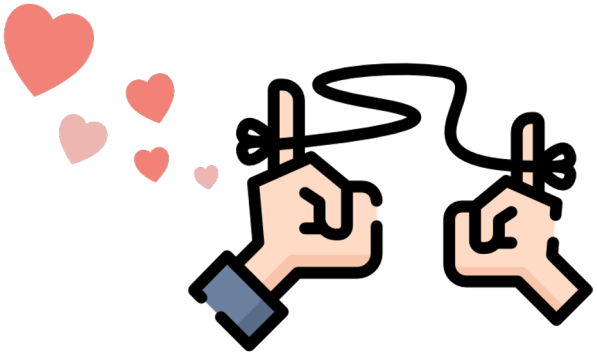
Manual interpretation cannot scale → we need AI that understands geography

Image sources: left, satellite tiles from CVDisaster (Li et al., 2025), highlighting is illustrative; middle, Yang et al. (2025), CEUS; right, Yang & Zou, When Satellite and Street Views Disagree (ACM SIGSPATIAL GeoSearch 2026).

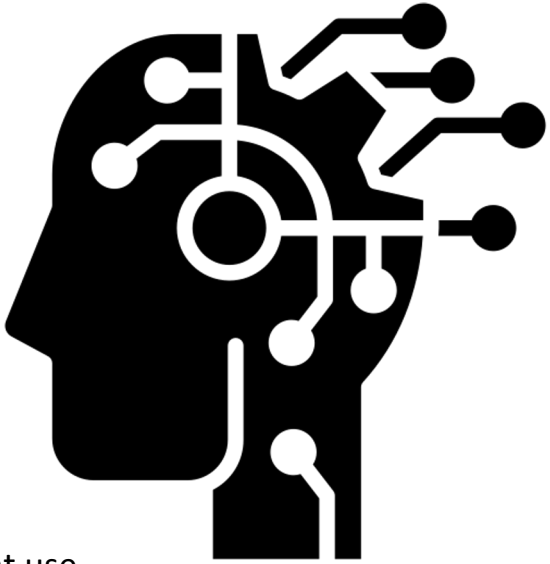
Background

GeoAI for disaster resilience

Geography



Artificial Intelligence



GeoAI refers to **spatially explicit AI** methods that use **geographic data and spatial context** to **create** intelligent geographic information, knowledge, and services.

Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. International Journal of Geographical Information Science, 34(4), 625–636.

Background

Three families of AI models, three roles in GeoAI

Discriminative models — task-specific classifiers

"Is this building damaged? Mild, moderate, or severe?"

+ Accurate on the task they are trained for

− Need labels for every event; stay confident even when wrong

Generative models — foundation and vision-language models

"Why is this location severely damaged? What should responders do here?"

+ Zero- and few-shot generalization across hazards and regions

− Usually less accurate than tuned classifiers; may hallucinate

Agentic AI = Large Language Models (LLMs) + tools + context

"Plan the analysis, call the spatial tools, and show the evidence behind every claim."

+ Combines both families: classifiers for accuracy, foundation models for reasoning

+ Calls GIS tools and live data, so answers rest on computed evidence, not memory

+ Extends beyond damage assessment to the whole resilience cycle

Bommasani, R., et al. (2021). On the opportunities and risks of foundation models. arXiv:2108.07258. · Wang, L., et al. (2024). A survey on large language model based autonomous agents. Frontiers of Computer Science, 18(6), 186345. · Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. ICML.

Background — one-sentence summary

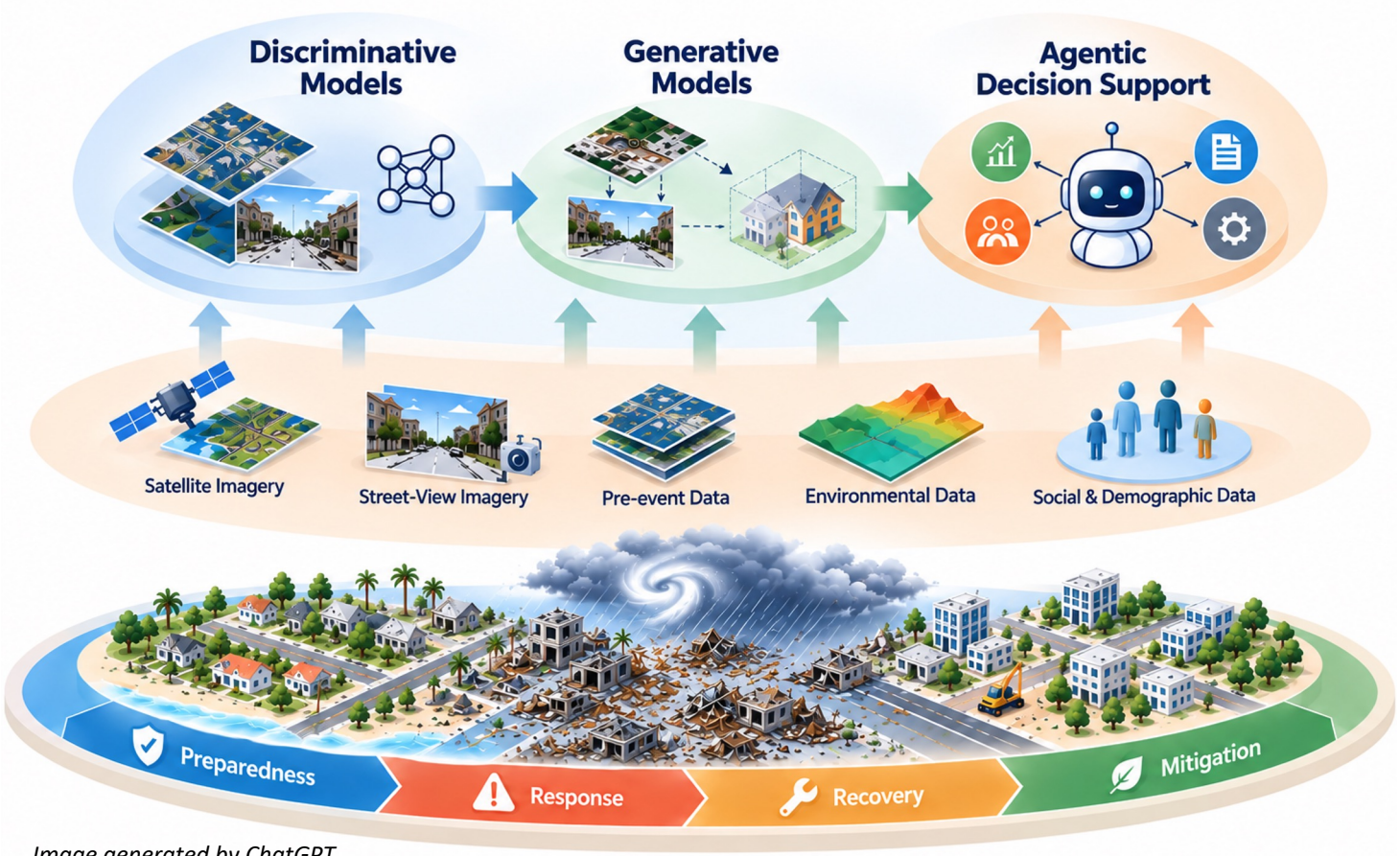


Image generated by ChatGPT

Advancing GeoAI for Disaster Resilience:
From Damage Assessment to Insightful Decision Support

Current Challenges

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1 **Discriminative assessment is not yet accurate or reliable enough.**

Models that see only one post-event view misclassify damage and stay confident when wrong.

How to see better?

2 **Specialized models do not generalize across events.**

Each new disaster needs new labels and retraining, which is too slow for response.

How to respond faster?

3 **Assessment alone does not close the resilience loop.**

A damage label or map is not a decision; each phase needs actionable, accountable guidance.

How to prepare and deliver better?

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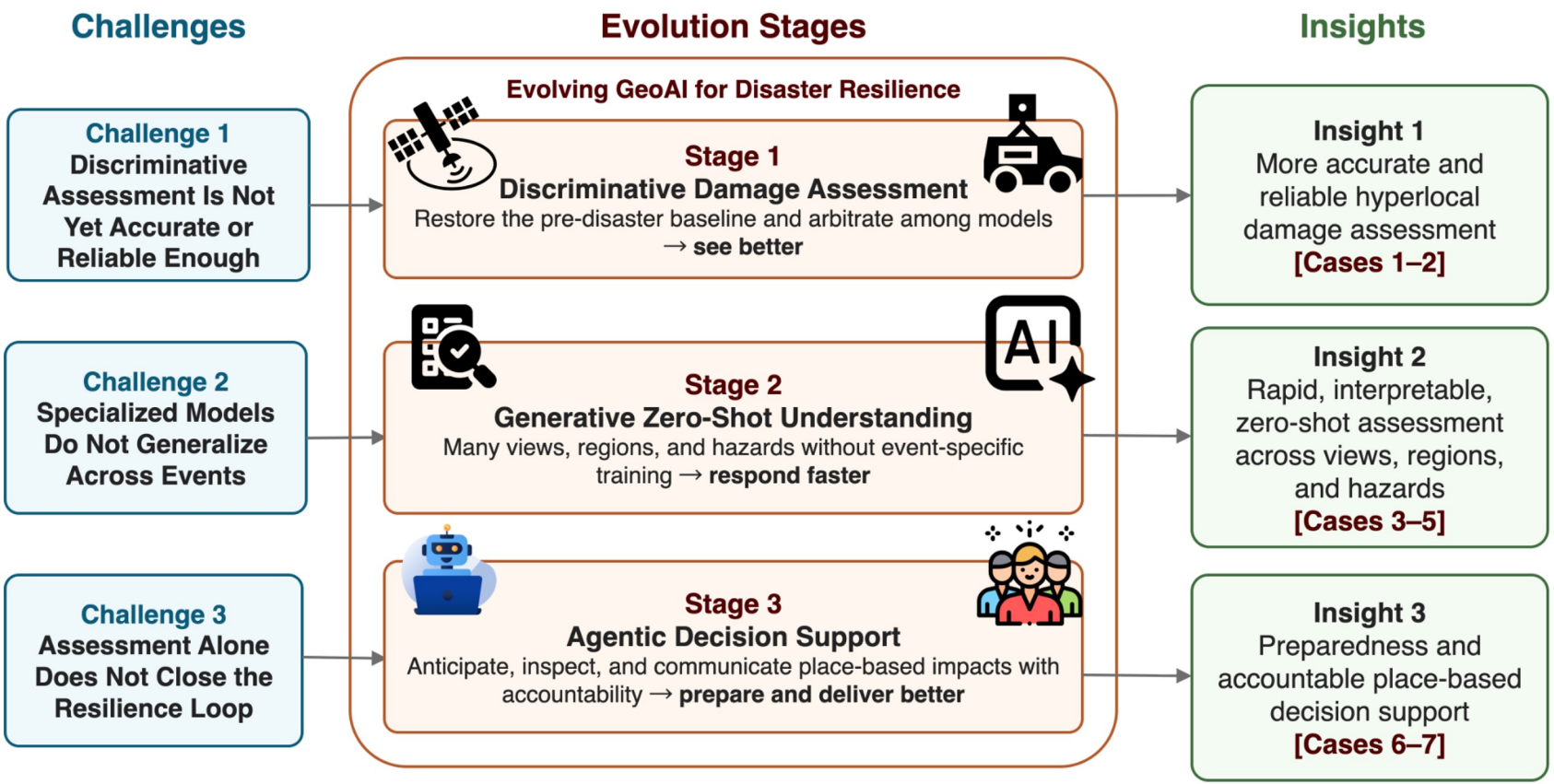
Conceptual Framework



STAGE 1 · OBJECTIVE 1

See better

Challenge 1 · Discriminative assessment is not yet accurate or reliable enough

Improve accuracy and reliability by fusing pre- and post-event views and arbitrating among multiple models.

Case 1	**Hyperlocal damage assessment from bi-temporal street-view imagery**	Published · CEUS 2025
Case 2	**DamageArbiter: multimodal arbitration for damage assessment**	Under review · Natural Hazards

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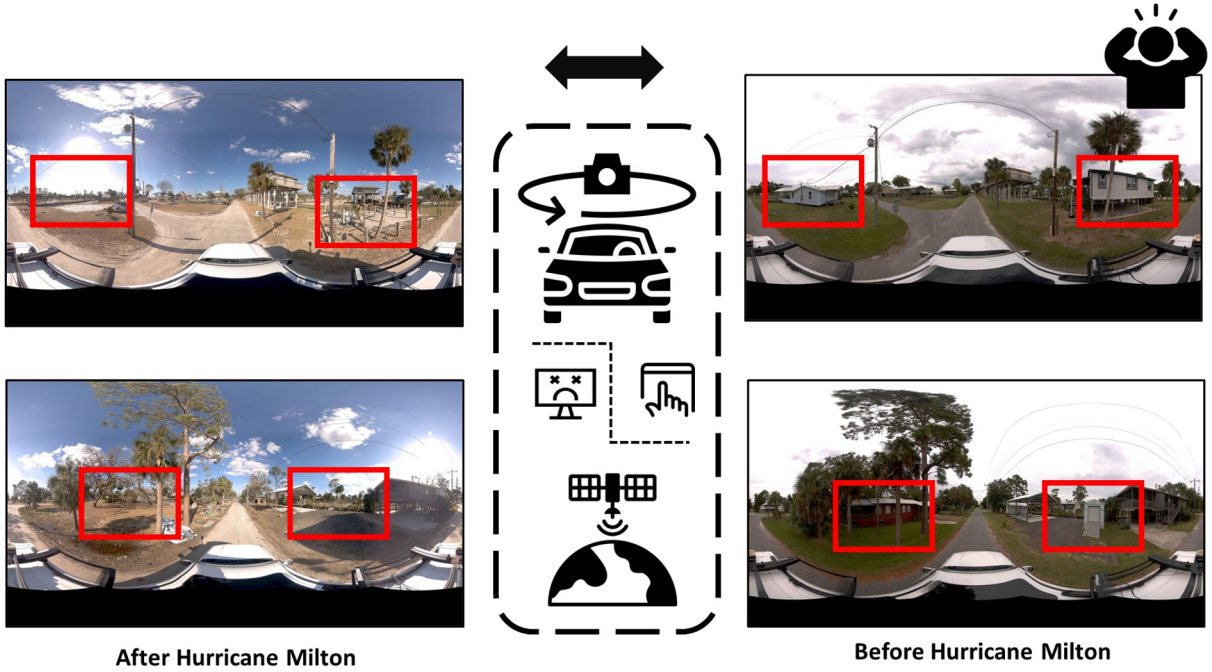
Case 1:

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)



After Hurricane Milton

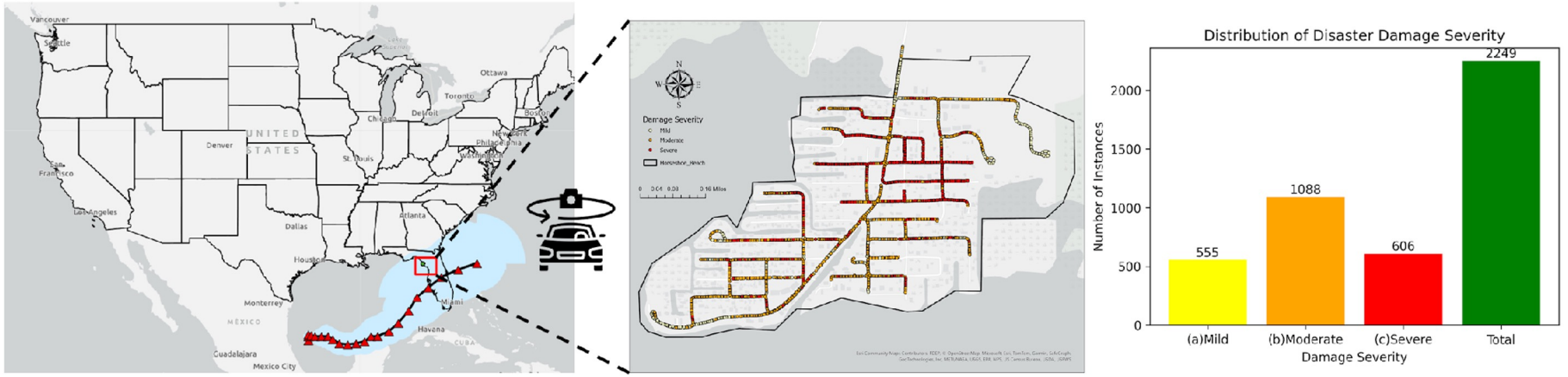
Before Hurricane Milton

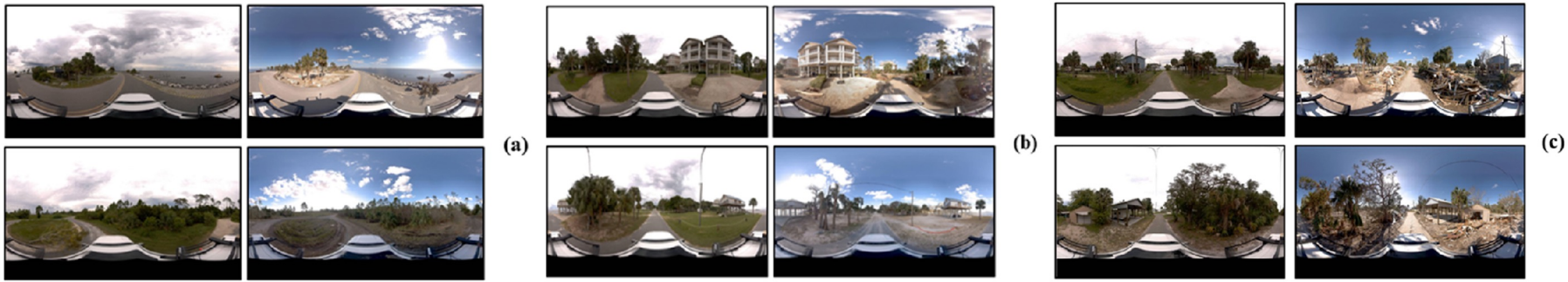
Research questions:

(1) Can pre-disaster street-view imagery help us classify disaster damage more accurately?

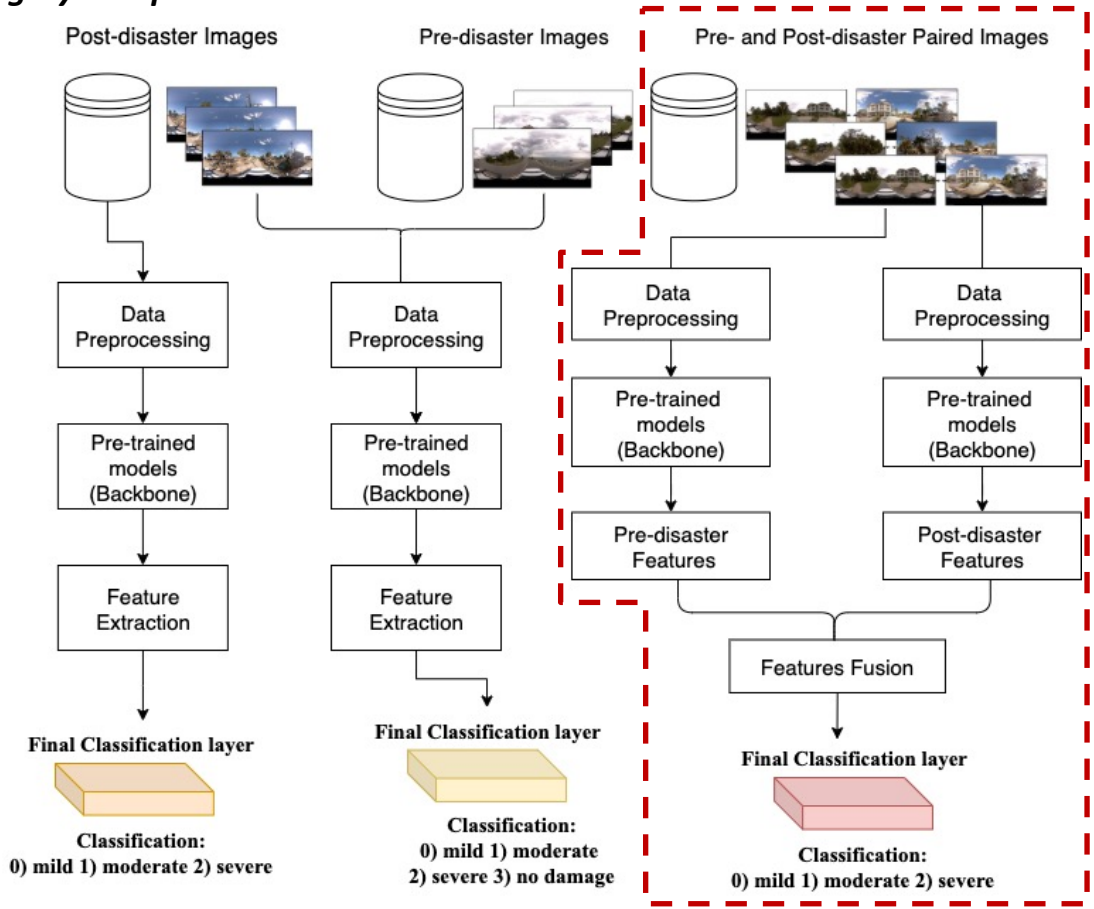
(2) How can we integrate pre- and post-disaster street-view imagery to enable rapid, hyperlocal damage assessment?

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (study area and dataset)





Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



Three settings:

(1) **Post-only**: classify post-event images into 3 levels

(2) **Pre- and Post-**: pre- and post-event images mixed together, 4 classes (+ no damage)

(3) **Pre- and Post-Paired**: pre- and post-event features fused

Next slide: how setting (3) extracts and fuses the two views

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

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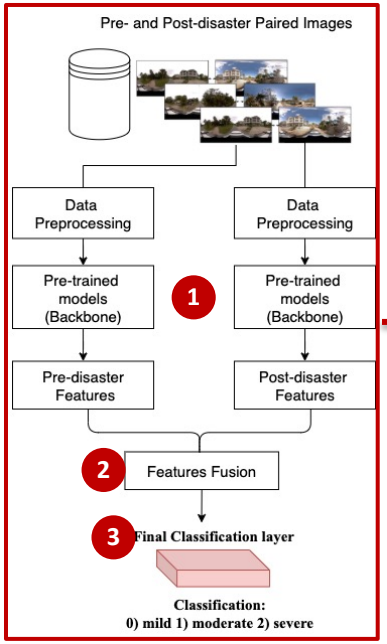
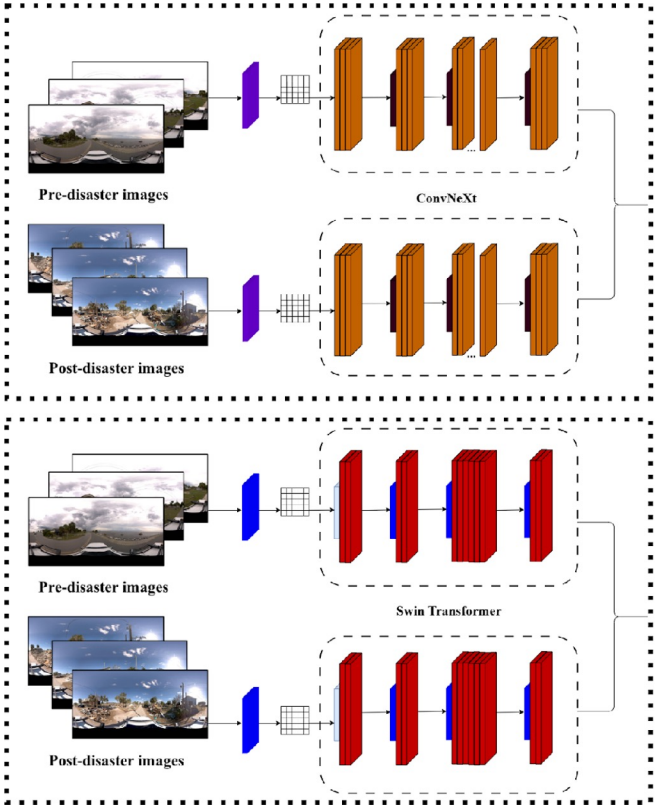
Setting (3), previous slide



The right side opens up steps 1 to 3

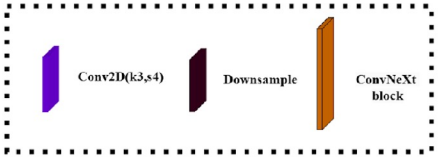
1 Two backbones: pre- and post-disaster features

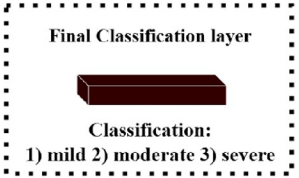


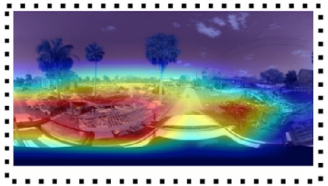
2 Feature fusion

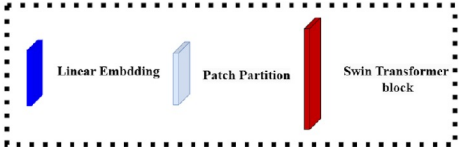
- 6-channel Baseline Swin Transformer
- OR
- 6-channel Baseline ConvNext
- OR
- Dual-Swin Transformer Cross-Attention
- OR
- Dual-ConvNeXt Cross-Attention
- OR
- Feature Fusion ConvNext (Weighted Fusion)
- OR
- Siamese ConvNext (Feature Difference)
- OR
- Dual-Swin Transformer

3 Classification and Grad-CAM

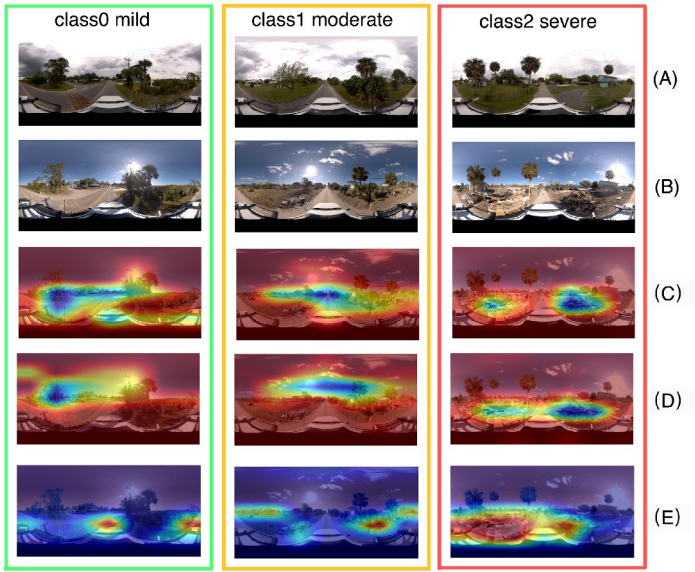




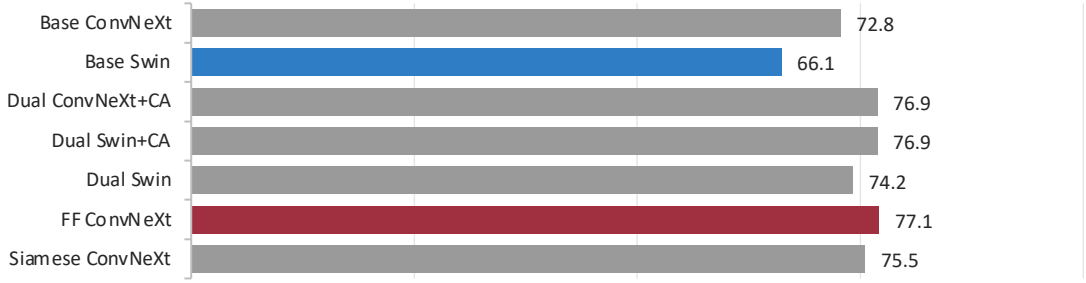




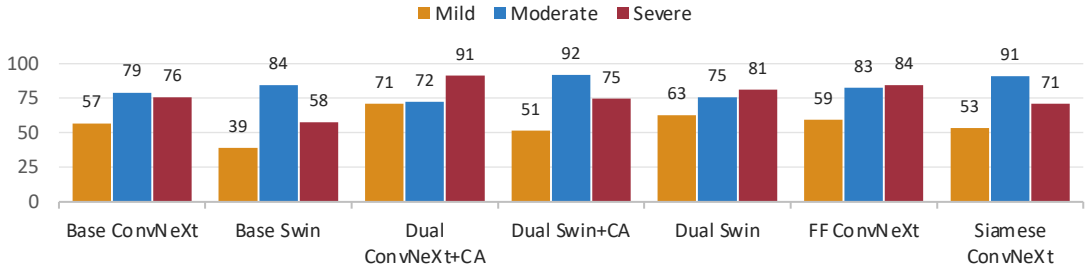
Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



Overall accuracy (%): single-view baselines vs. dual-channel models



Class-specific accuracy (%) · CA = cross-attention, FF = feature fusion



Key result: Accuracy improved from 66.14% (Swin baseline) to 77.11% (dual-channel Feature-Fusion ConvNeXt); Grad-CAM confirms pre-disaster images help focus on true pre/post changes.

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)

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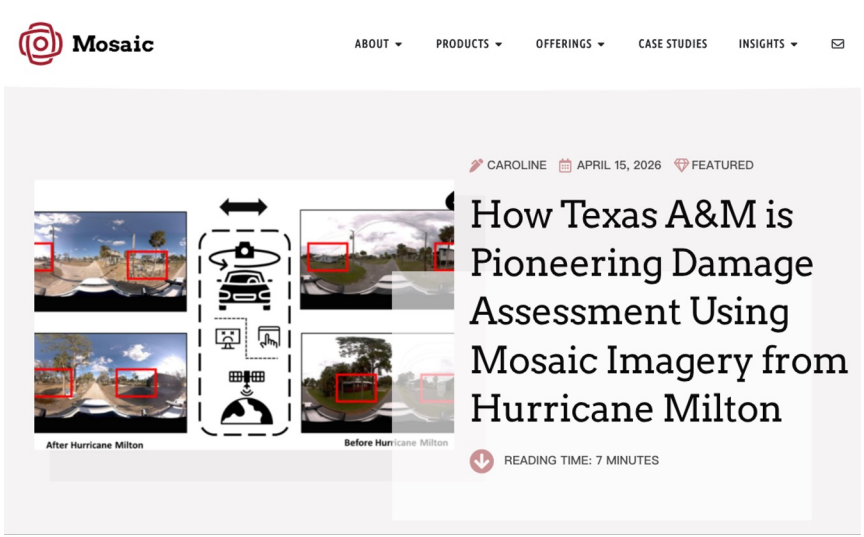

https://www.mosaic51.com/featured/how-texas-am-is-pioneering-damage-assessment-using-mosaic-imagery-from-hurricane-milton/

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Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

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Dual-channel neural network
Pre-trained vision model
Damage estimation

ABSTRACT

Street-view images offer unique advantages for disaster damage estimation as they capture impacts from a visual perspective and provide detailed, on-the-ground insights. Despite several investigations attempting to analyze street-view images for damage estimation, they mainly focus on using post-disaster images. The potential of time-series street-view images remains underexplored. Pre-disaster images provide valuable benchmarks for accurate damage estimations at building and street levels. These images could also aid annotators in objectively labeling post-disaster impacts, improving the reliability of labeled data sets for model training, and potentially enhancing the model performance in damage evaluation. The goal of this study is to estimate hyperlocal, on-the-ground disaster damages using bi-temporal street-view images and advanced pre-trained vision models. Street-view images before and after 2024 Hurricane Milton in Horseshoe Beach, Florida, were collected for experiments. The objectives are: (1) to assess the performance gains of incorporating pre-disaster street-view images as a no-damage category in fine-tuning pre-trained models, including Swin Transformer and ConvNeXt, for damage level classification; (2) to design and evaluate a dual-channel algorithm that reads pair-wise pre- and post-disaster street-view images for hyperlocal damage assessment. The results indicate that incorporating pre-disaster street-view images and employing a dual-channel processing framework can significantly enhance damage assessment accuracy. The accuracy improves from 66.14 % with the Swin Transformer baseline to 77.11 % with the dual-channel Feature-Fusion ConvNeXt model. Gradient-weighted Class Activation Mapping (Grad-CAM) shows that incorporating pre-disaster images improves the pre-trained vision model's capacity to focus on major changes between pre- and post-disaster images, thus enhancing model performance. This research evaluates the technical solutions and challenges of assessing disaster damages at hyperlocal spatial resolutions with bi-temporal street-view images, providing valuable insights to support effective decision-making in disaster management and resilience planning.

Yang, Y., Zou, L., Zhou, B., Li, D., Lin, B., Abedin, J., & Yang, M. (2025). Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models. Computers, Environment and Urban Systems, 121, 102335.

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (Under Review, Natural Hazards)

Post-Hurricane Street-View Imagery

Image-only model (e.g., ViT-B/32)

Mild

(mild 70%, moderate 20%, severe 10%)

Actual value: Severe

Overconfident error: the model is wrong, yet sure of its answer.

Confidence gap C = p(predicted class) − p(true class) = 0.70 − 0.10 = 0.60

C = 0.60 ≥ 0.40 → this error counts as overconfident

Image-only is not enough

- The model sees only pixels
- It answers “mild”, 70% sure
- The truth is severe: wrong and overconfident
- Crews can be sent to the wrong place

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Image-to-text: an LLM describes each image and names the damage cues

Mild

LLM describes

Sagging power lines are visible alongside a clear, dry road. Many trees show **defoliation** and some **branch damage**, but no large fallen trees or building debris are present. The overall scene suggests **mild environmental impact**, with infrastructure largely intact.

Moderate

LLM describes

Scattered wooden debris and **loose utility lines** on a power pole indicate **moderate damage** along this coastal dirt road. While buildings appear structurally sound, the presence of **scattered planks and beams**, alongside the compromised power infrastructure, suggests the area experienced some impact, but not widespread destruction or flooding.

Severe

LLM describes

The street is heavily impacted by **extensive building debris**, forming **large piles** along both sides, almost completely **blocking passage**. This significant accumulation of scattered materials and **collapsed fragments** suggests **severe damage** to structures in the vicinity, though visible infrastructure like power lines appears mostly intact.

Images alone are not enough, so we add text.

RQ1: Beyond accuracy, how reliable are image-only, text-only, and image-text models?

RQ2: Can arbitration between models reduce overconfident errors while keeping accuracy?

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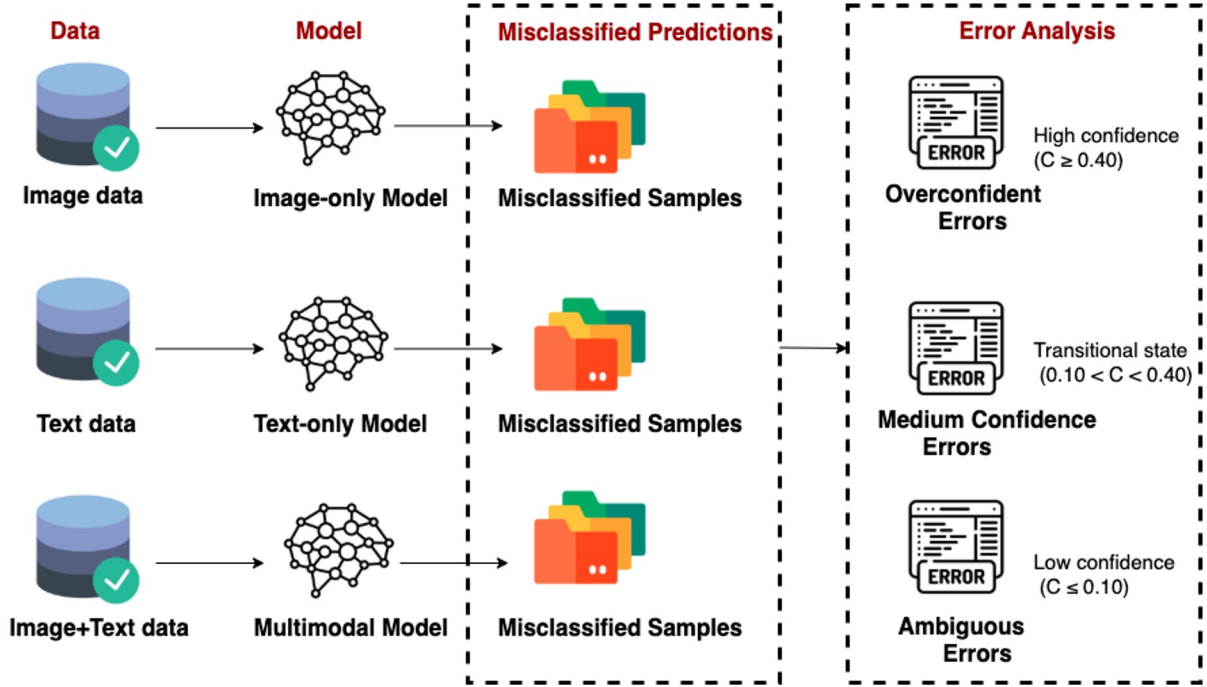

Every error is sorted by its confidence gap C

C = p(predicted class) − p(true class)

Earlier example: C = 0.70 − 0.10 = 0.60, so it falls in the top group (C ≥ 0.40).

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Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

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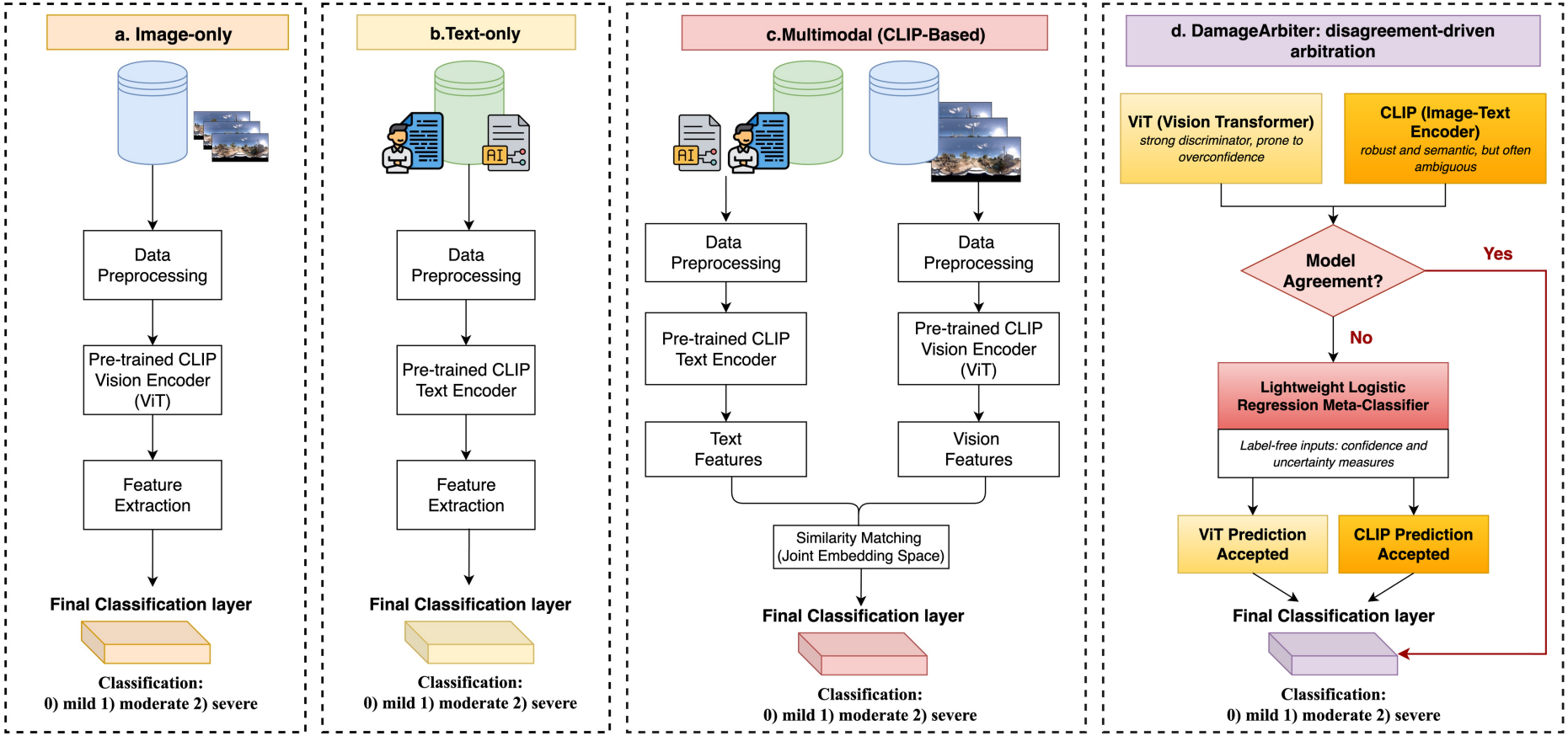

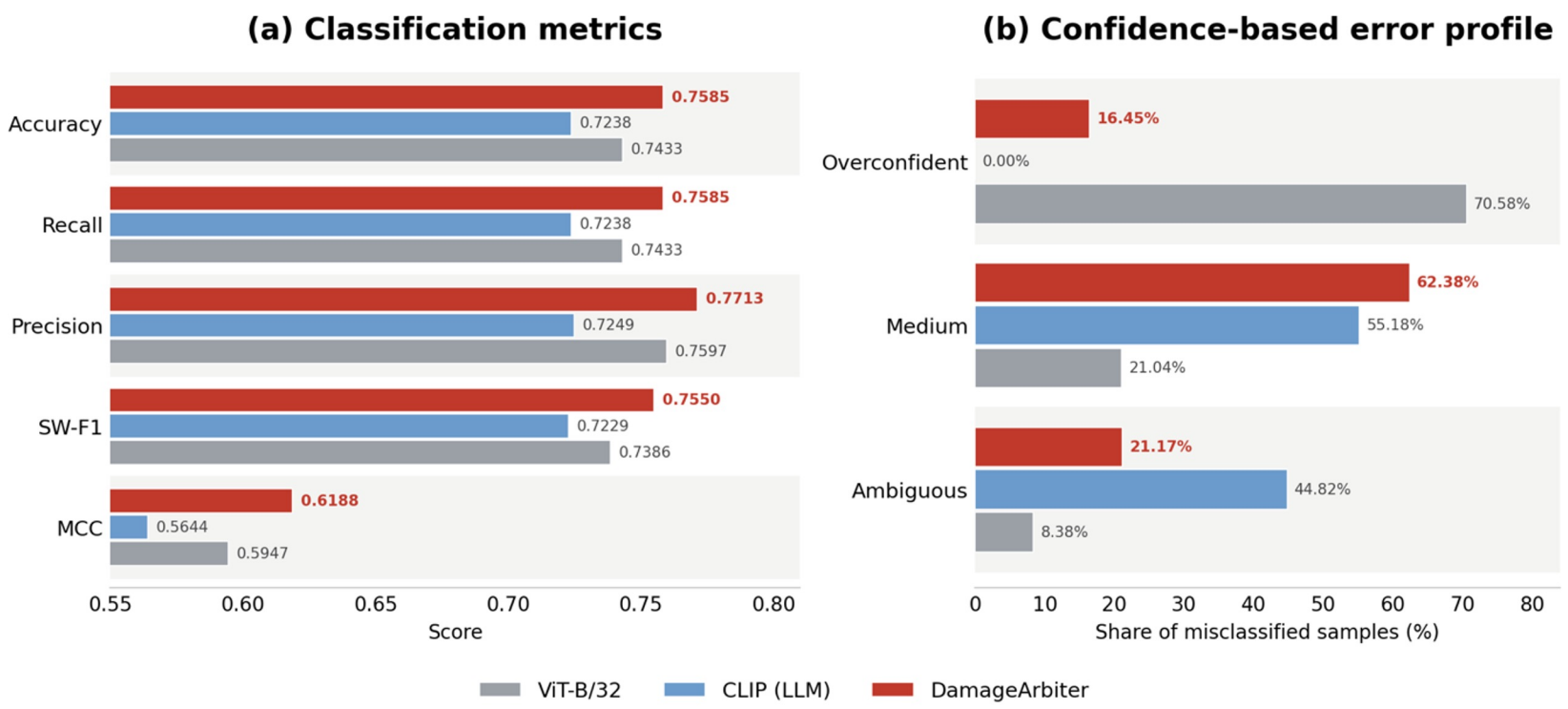
(a) Classification metrics
Accuracy
0.7585
0.7238
0.7433
Recall
0.7585
0.7238
0.7433
Precision
0.7713
0.7249
0.7597
SW-F1
0.7550
0.7229
0.7386
MCC
0.6188
0.5644
0.5947
0.55
0.60
0.65
0.70
0.75
0.80
Score
(b) Confidence-based error profile
Overconfident
16.45%
0.00%
70.58%
Medium
62.38%
55.18%
21.04%
Ambiguous
21.17%
44.82%
8.38%
0
10
20
30
40
50
60
70
80
Share of misclassified samples (%)
ViT-B/32
CLIP (LLM)
DamageArbiter

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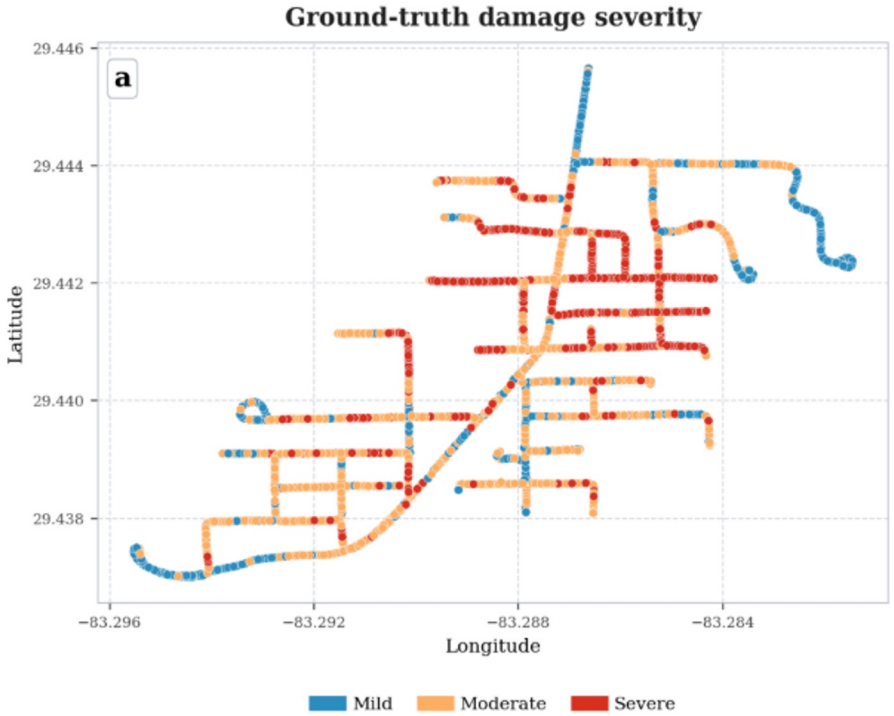

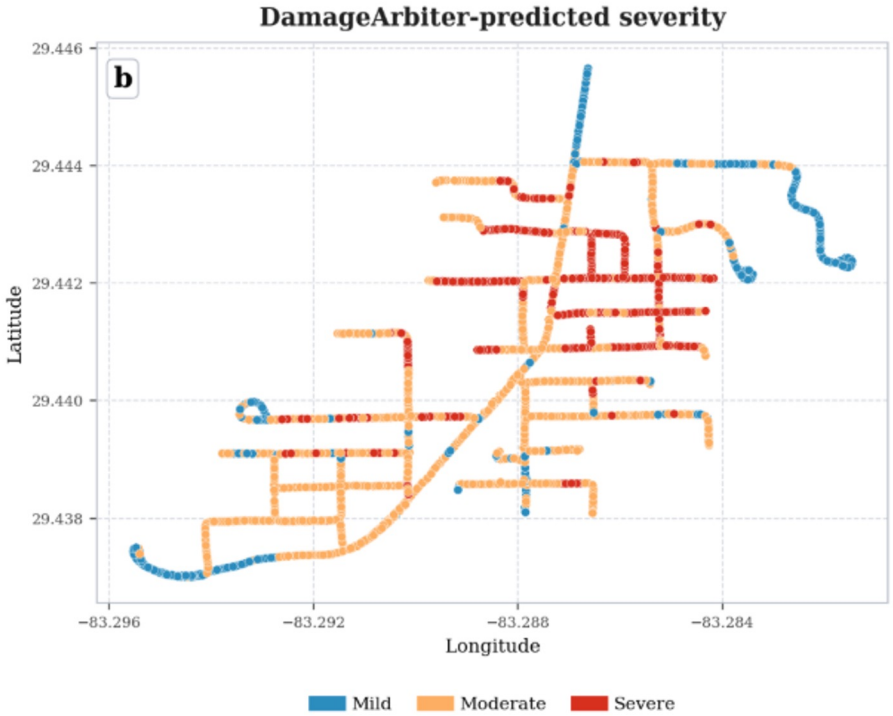

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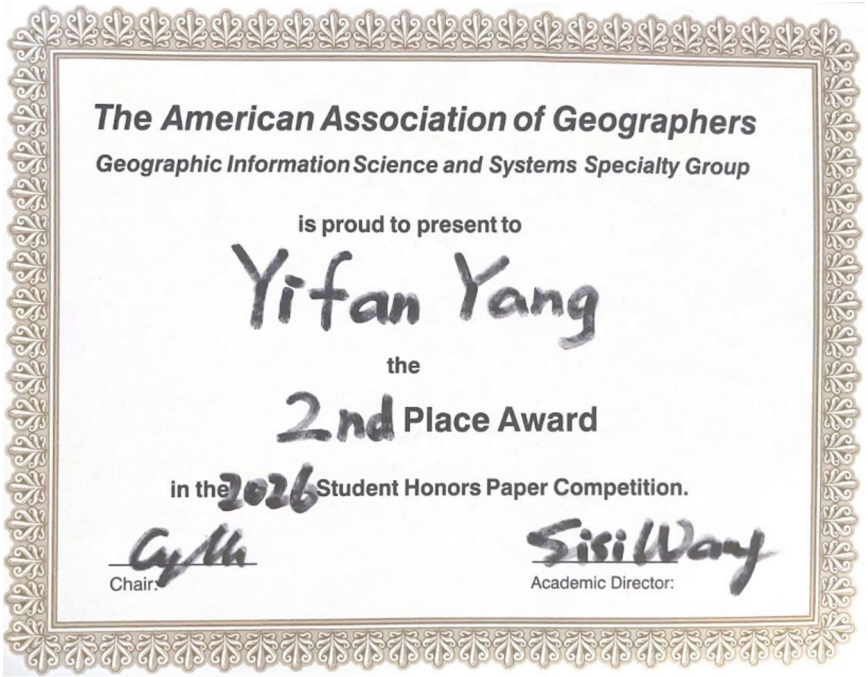
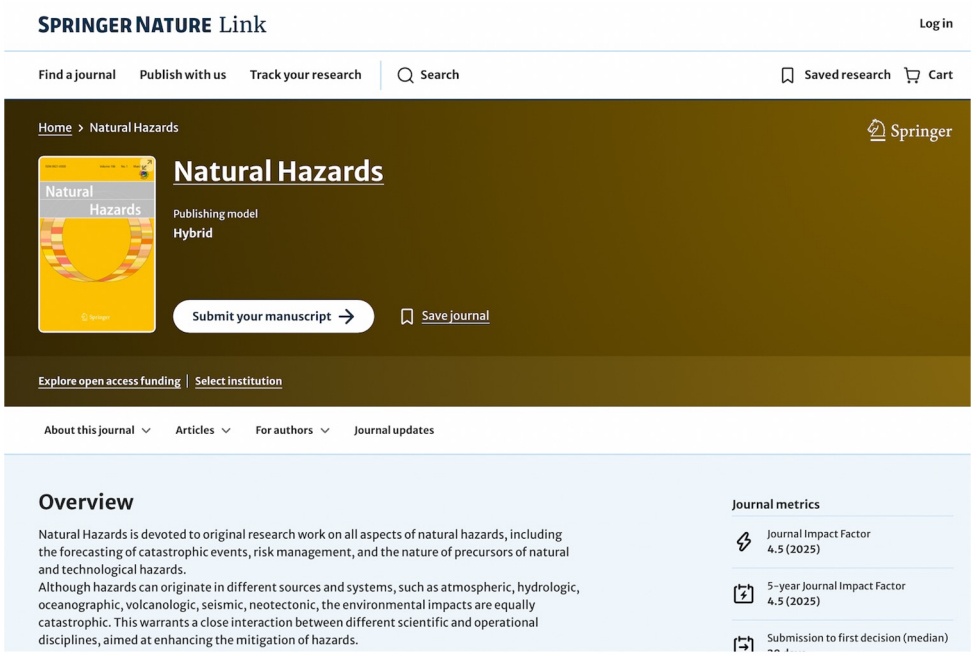
Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (Under Review, Natural Hazards)





Yang, Y., Zou, L., Gong, W., Fu, K., Li, Z., Wang, S., ... & Tian, H. (2026). DamageArbiter: A CLIP-enhanced multi-modal arbitration framework for hurricane damage assessment from street-view imagery. arXiv preprint arXiv:2603.14837.

STAGE 2 · OBJECTIVE 2

Respond faster

Challenge 2 · Specialized models do not generalize across events

Enable zero-shot assessment that transfers across hazards and regions without new labels or retraining.

Case 3	**RAPID: multi-agent pipeline for interpretable cross-view assessment**	Accepted · ACM SIGSPATIAL 2026
Case 4	Satellite-to-Street: post-disaster street views from satellite imagery	Accepted · IEEE IGARSS 2026
Case 5	Generative AI for disaster resilience	Under review · CRC Press

Focus today: Case 3. More Case 3 results and Cases 4, 5, and 8 are in the backup slides.

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Case 3:

RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (Accepted, ACM SIGSPATIAL 2026)

Research Question

Can RAPID support:

- Cross-view assessment?
- Different regions and hazards?
- Rapid zero-shot reasoning?
- Interpretable damage reports?

Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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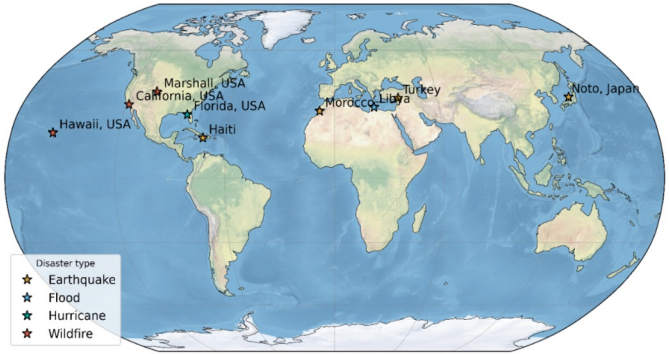
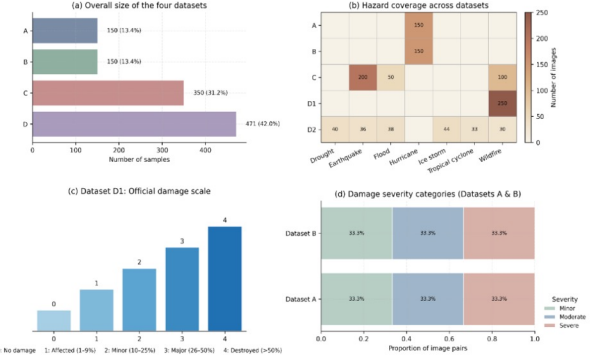
Stage 1. Data Preparation

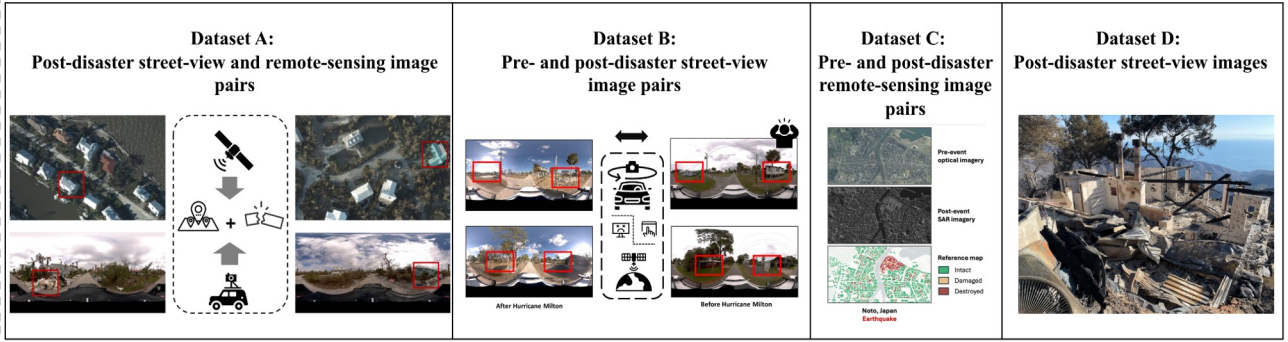
a. Geographic location of events



b. Key statistics



c. Dataset description



a. Events

- 9 sites, 4 continents: USA ×4, Haiti, Morocco, Libya, Turkey, Japan
- Earthquake, flood, hurricane, wildfire (+ ice storm, drought in D2)

b. Key statistics

- 4 datasets, ~1,100 images (A 150 · B 150 · C 350 · D 471)
- Balanced severity in A / B; official 0–4 scale in D1

c. Datasets

- A: post-event street view + satellite (CVDisaster)
- B: pre/post street view (BiTemporal StreetView)
- C: pre/post satellite + building masks (BRIGHT)
- D: post-event street view, geo-referenced or web-sourced

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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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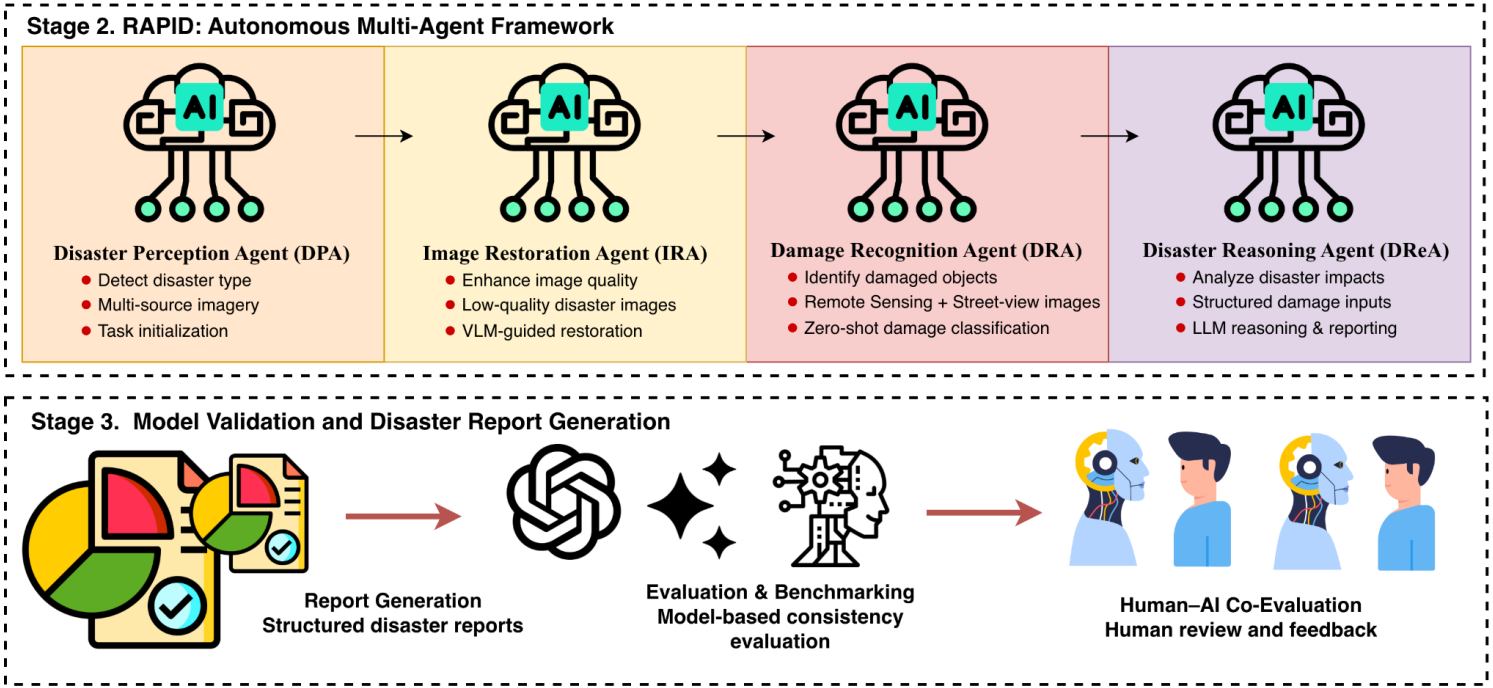

What I did				
	Workflow design Four agents, each with one clear task	**Prompt engineering** Zero-shot prompts for every agent; no training	**Benchmark data** Four datasets, ~1,100 images, two views	**Validation** Three models compared; model and human review

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Disaster Perception Agent (DPA): hazard recognition is the most reliable stage

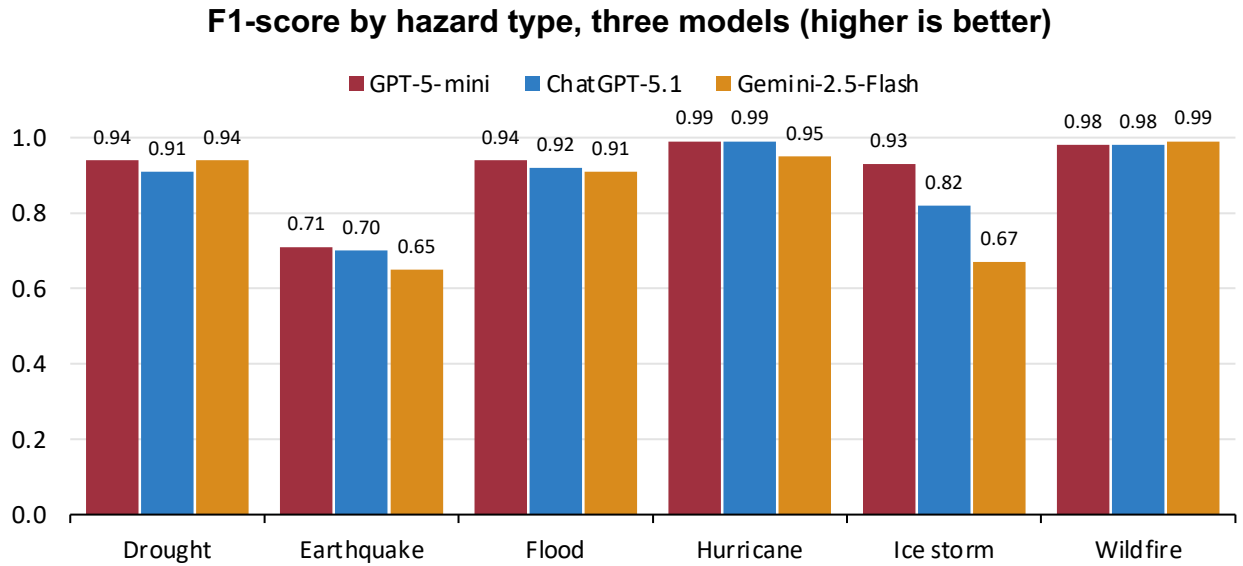
What it does

- Identifies the input: satellite, street view, or a pair
- Classifies one of six hazards
- Flags images for restoration and plans the route

0.92

overall accuracy (GPT-5-mini)

ChatGPT-5.1: 0.88 · Gemini-2.5-Flash: 0.86



F1 ≥ 0.91 on drought, flood, hurricane, and wildfire for every model. Weak spot: earthquake, where recall is only 0.50 to 0.58.

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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

Agent 2 of 4 · Restoration (IRA)

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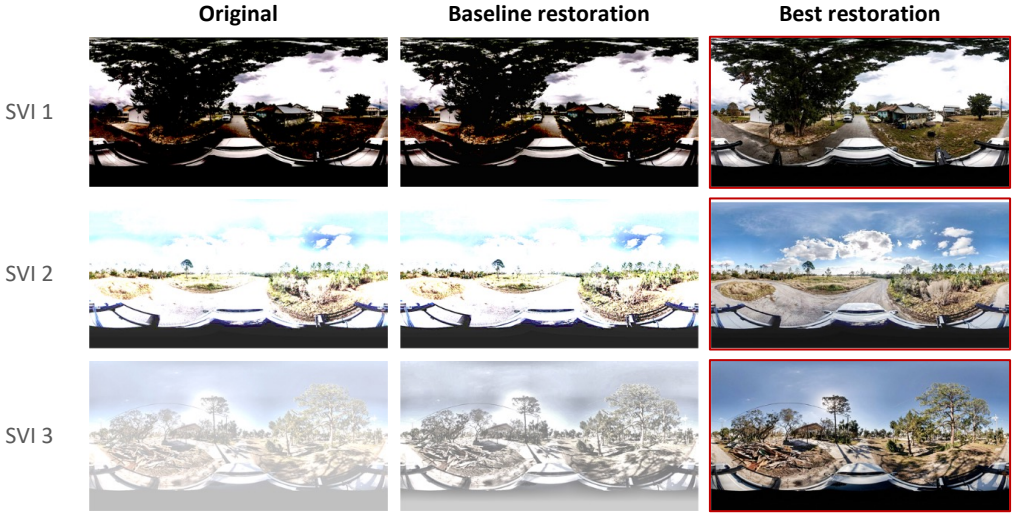
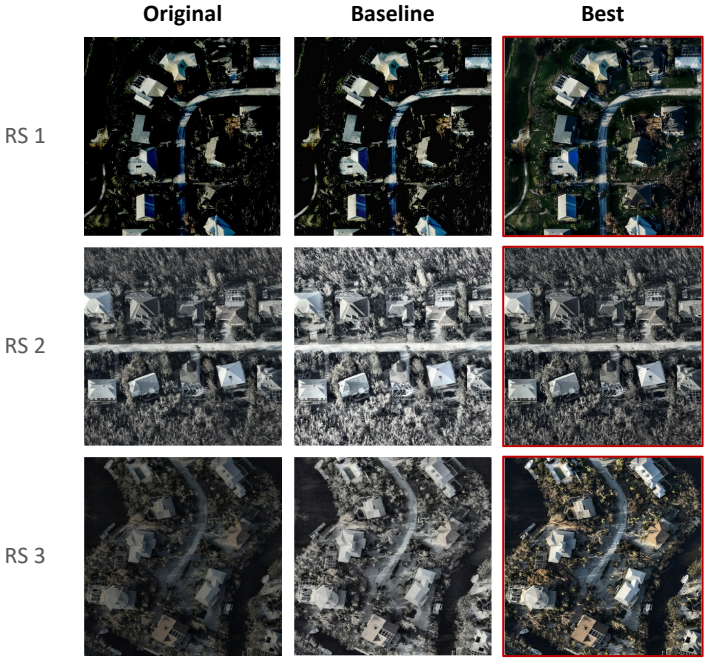
Street-view imagery



Satellite / aerial imagery



Where we are in RAPID: agent 2 of 4

DPA → **IRA** → DRA → DReA

Red outline: the best restored result. The original image always stays available for audit.
Quality scores (Q) for every hazard are in the backup slides.

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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

Agent 3 of 4 · Recognition (DRA)

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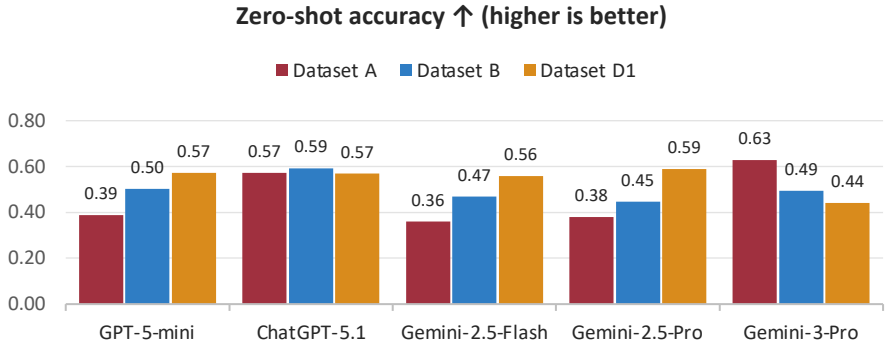

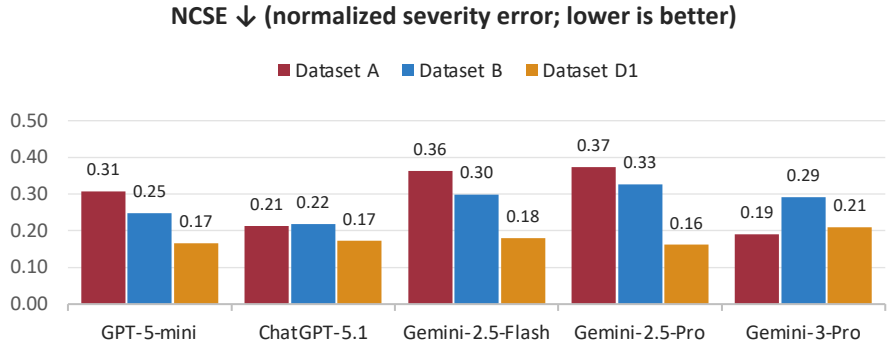

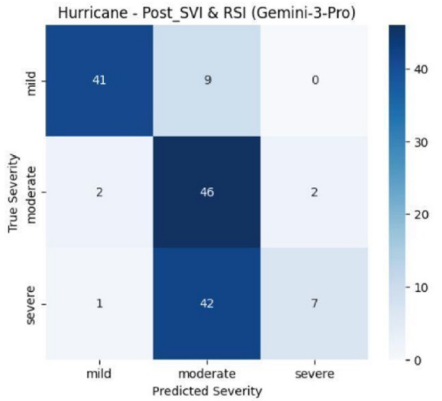

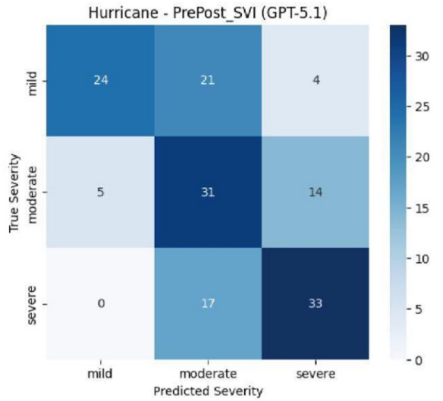

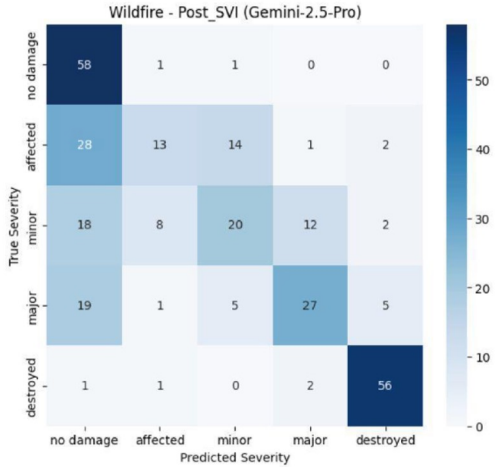

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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

Agent 4 of 4 · Reasoning (DReA)

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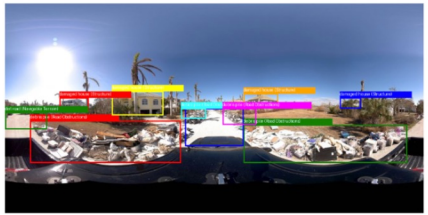
Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building
 Using Gemini-3-Pro for object recognition	The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. **Reasoning for Gemini-3-Pro-Disaster Assessment**			
	Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. **Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion**			

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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (Accepted, ACM SIGSPATIAL 2026)

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RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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Abstract

Due to the increasing frequency and intensity of climate extreme events, there is a clear demand for intelligent, scalable, and autonomous approaches to disaster damage assessment. Existing methods, largely based on supervised learning and task-specific fine-tuning strategies, struggle to generalize under domain shifts, long-tailed data distributions, and heterogeneous geospatial data sources, especially in disaster scenarios. In addition, they often lack the ability to integrate and reason effectively across multimodal geospatial information, such as satellite images and street-view images. In this paper, we introduce RAPID, a reproducible multi-agent pipeline for interpretable disaster damage assessment, encompassing the assessment of damage levels, interpretation of damage types and degrees, and generation of actionable suggestions for response, remediation, and recovery. RAPID coordinates multiple specialized agents to perform cross-view understanding, image restoration, structured damage recognition, and geographical reasoning across heterogeneous data modalities. Without task-specific fine-tuning, RAPID supports zero-shot damage assessment by jointly leveraging complementary information from remote sensing and ground-level perspectives. Furthermore, the system produces fine-grained, interpretable damage assessments and automatically generates location-specific, decision-relevant disaster reports to support early-stage emergency response. We evaluated and demonstrated the performance of the proposed RAPID framework across multiple disaster using different sets of cross-view imagery inputs, including street-view images before and after the disaster, remote sensing imagery after the disaster, and street-view image pairs. Experiments show that RAPID achieves an overall accuracy of 0.92 for multi-disaster type classification and up to 0.627 for cross-view damage severity prediction, highlighting its potential as a foundational framework for autonomous disaster intelligence.

CCS Concepts

• **Information systems** → **Geographic information systems**; • **Computing methodologies** → *Artificial intelligence*; *Computer vision tasks*.

Keywords

Disaster damage assessment, vision-language models, cross-view imagery, zero-shot learning, multi-agent pipeline

ACM Reference Format:

Yifan Yang, Wenjing Gong, Kaili Zhang, Lei Zou, Zhengzhong Tu, Hao Li, Zongrong Li, and Xinyue Ye. 2026. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. In *The 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26), November 03–06, 2026, Riverside, CA, USA.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3841645.3843346

Yifan Yang, Wenjing Gong, Kaili Zhang, Lei Zou, Zhengzhong Tu, Hao Li, Zongrong Li, and Xinyue Ye. 2026. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. In The 34th ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26), November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3841645.3843346

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STAGE 3 · OBJECTIVE 3

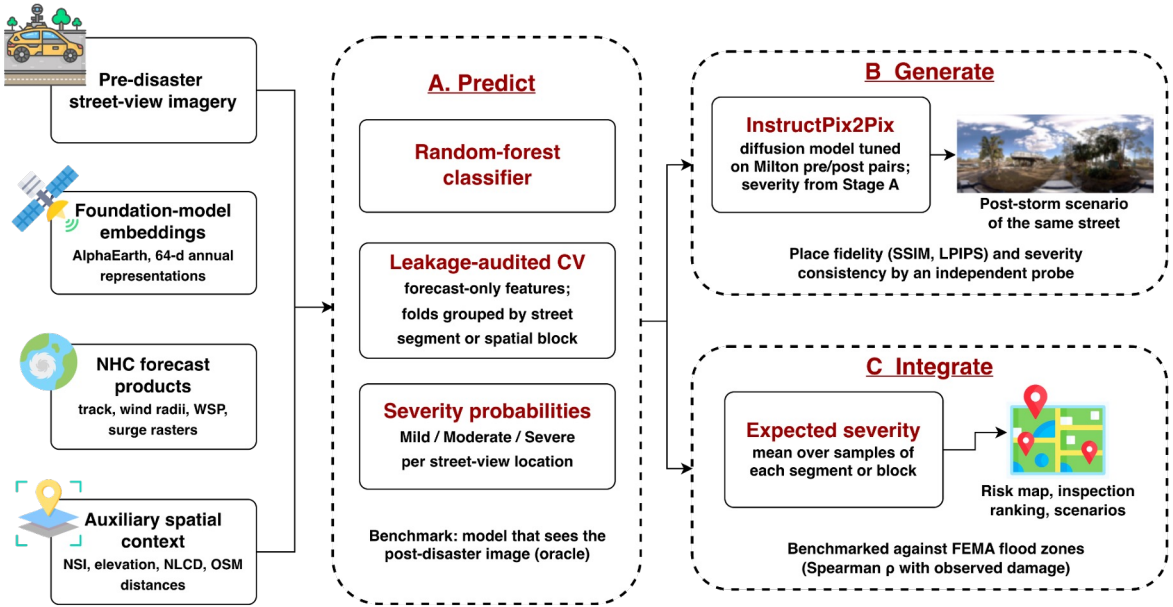
Prepare and deliver better

Challenge 3 · Assessment alone does not close the resilience loop

Build accountable agentic GeoAI that turns assessment into evidence-backed action before, during, and after disasters.

Case 6	**Seeing damage before it happens: anticipatory street-level prediction**	Ongoing
Case 7	**RAY: an accountable GeoAI system for place-based disaster intelligence**	Ongoing

Case 6: Seeing Disaster Damage Before It Happens: Predicting and Visualizing Street-Level Impacts from Multimodal Pre-Event Information (Ongoing)



Case studies: Hurricane Milton 2024, Horseshoe Beach: Stages A, B, C Hurricane Ian 2022, Lee County: Stages A, C

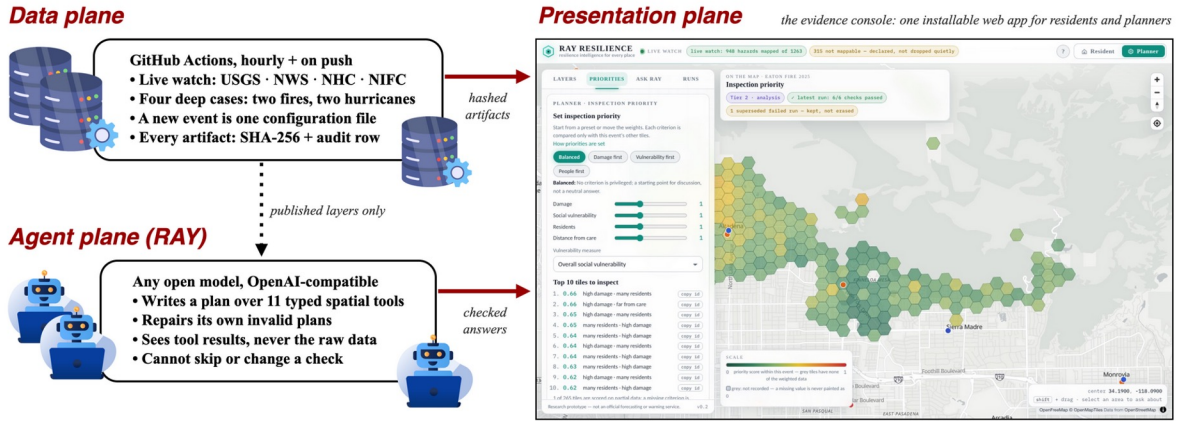
Research Question:

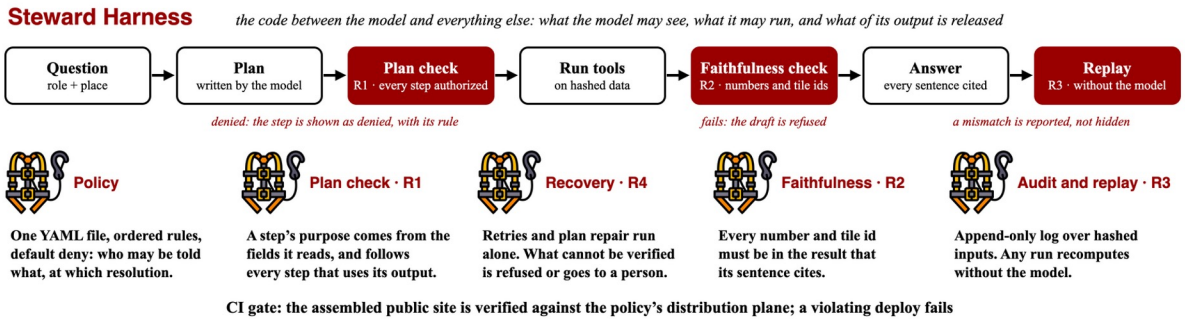
- **Can we anticipate post-disaster street-level damage using pre-disaster multi-modal data, Earth observation embeddings, and hazard forecasts?**

Objective:

- **To develop a preparedness-oriented GeoAI framework that transforms pre-disaster data and forecasts into anticipatory damage predictions, spatial risk prioritization, and scenario visualizations.**

Case 7: RAY: Resilience Assistant for You
An Accountable GeoAI System for Place-Based Disaster Intelligence (Ongoing)





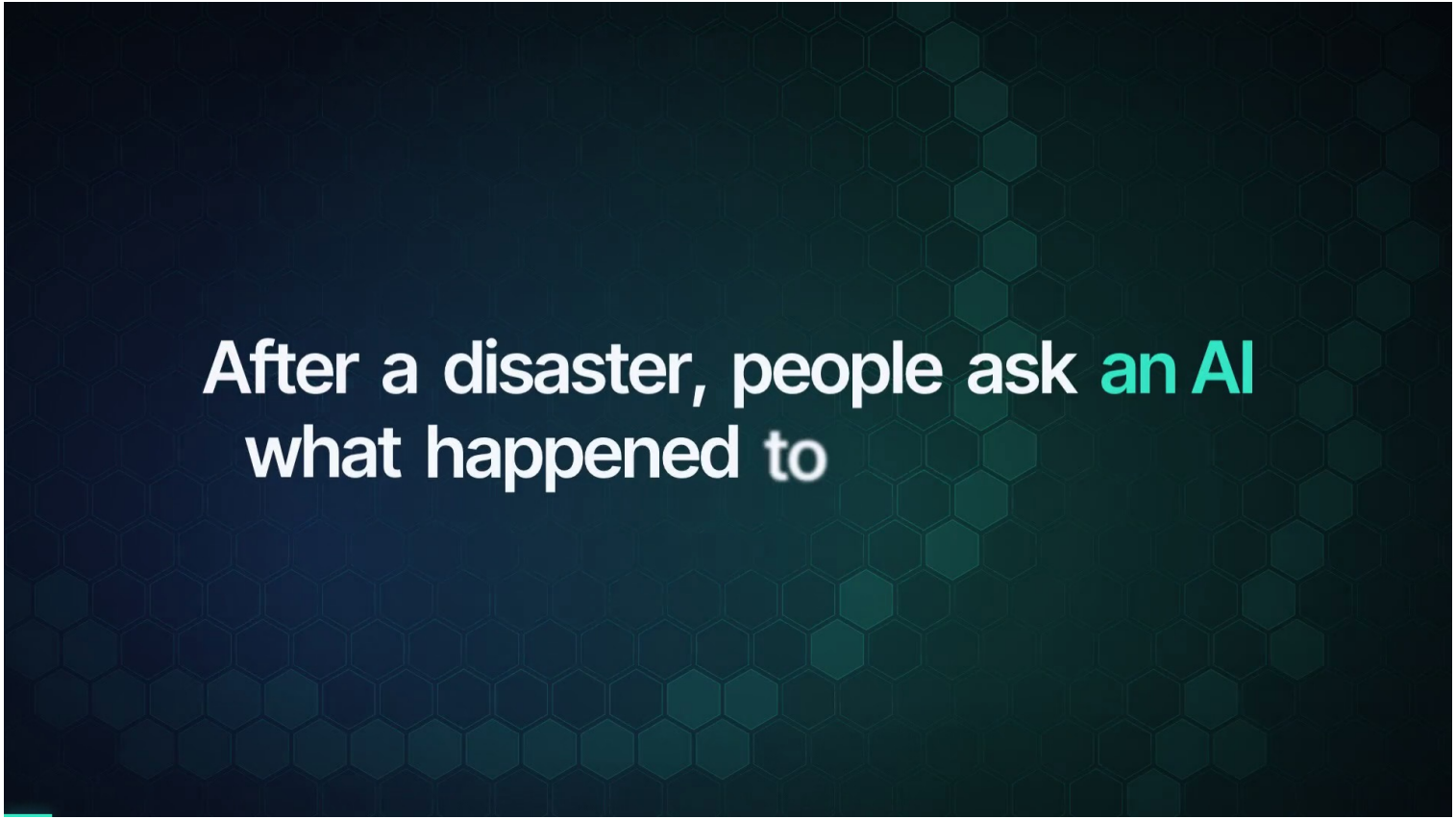
Research Question:

How can a GeoAI agent analyze disasters on its own, yet have every answer authorized, evidence-faithful, and reproducible, enforced by the system rather than promised by the model?

Objective:

Build RAY: an agent wrapped in a Steward Harness (policy check → grounded analysis → citation check → audit & replay), delivered through a WebGIS / mobile console for planners and residents.

Yifan Yang | Ph.D. Student @ Texas A&M University | Email: yyang295@tamu.edu | Lab Website: https://www.geoearlab.com



Link to the one-minute demo video:
https://rayford295.github.io/ray-resilience/demo/ray-resilience-teaser.mp4

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Publications and Awards

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As of November 2026 · Google Scholar: 442 citations, h-index 8, i10-index 7.

PUBLICATIONS

8 first-authored publications during Ph.D.

- 2 journal articles (CEUS 2025; Natural Hazards under review)
- 5 conference proceedings (ACM SIGSPATIAL 2026; ACM SIGSPATIAL Workshop GeoSearch '26; IEEE IGARSS 2026; CaGIS 2026; ICC 2025)
- 1 chapter under review (CRC Press)

14 co-authored publications

- 8 journal articles (e.g., International Journal of Geographical Information Science, Transactions in GIS, Int. J. Applied Earth Observation and Geoinformation, Urban Informatics, Journal of Transport Geography, etc.)
- 3 conference papers (ACM SIGSPATIAL 2026 short paper; GeoAI 2026; AGU 2025) and 3 preprints

HONORS AND AWARDS

A total of 15 honors received during Ph.D.

2026

- 2nd Place, AAG GISSG Student Honors Paper Competition
- AAG International Geographic Information Fund (IGIF) Scholarship
- CaGIS Doctoral Scholarship Award
- IGARSS 2026 Travel Grant
- NSF I-GUIDE Summer School Travel Award
- ACM SIGSPATIAL 2026 Travel Award
- Geography Graduate Student Enhancement Scholarship, Department of Geography, etc.

2025

- Best Student Paper Award, International Cartographic Conference (ICC 2025)
- Environment and Sustainability Graduate Fellow Award, TAMU
- TAMIDS Student Ambassador Scholarship
- CaGIS International Travel Grant, etc.

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Research Schedule

Now

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Tasks	Fall 2024	Spring 2025	Summer 2025	Fall 2025	Spring 2026	Summer 2026	Fall 2026	Spring 2027	Summer 2027
Case Study 1	Done								
Case Study 2		Done							
Case Study 3				Done					
Case Study 4					Done				
Case Study 5					Done				
Case Study 6							In Progress (90%)		
Case Study 7							In Progress (65%)		
Prelim Exam						Done			
Proposal Defense							In Progress		
Dissertation Writing							In Progress		
Final Defense & Graduation									Planned

Done In Progress Planned

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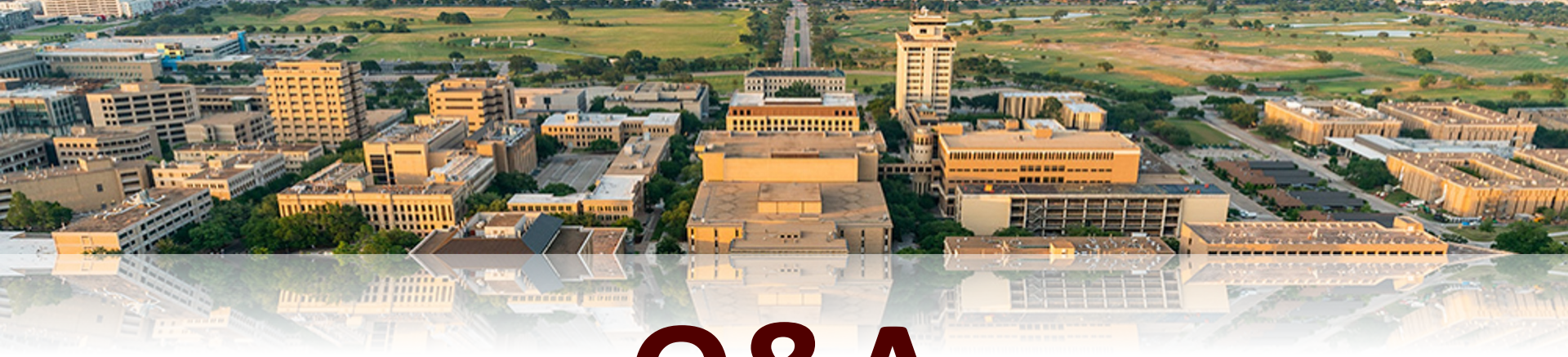

Q&A

Welcome any comments/questions

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See better · Respond faster · Prepare and deliver better

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Backup Slides

Additional results and cases, kept for questions

Case 3	**RAPID: restoration quality scores and reasoning evaluation**	**Accepted · ACM SIGSPATIAL 2026**
Case 4	Satellite-to-Street: post-disaster street views from satellite imagery	**Accepted · IEEE IGARSS 2026**
Case 5	Generative AI for disaster resilience	**Under review · CRC Press**
Case 8	When satellite and street views disagree: cross-view complementarity	**Accepted · GeoSearch 2026**

Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

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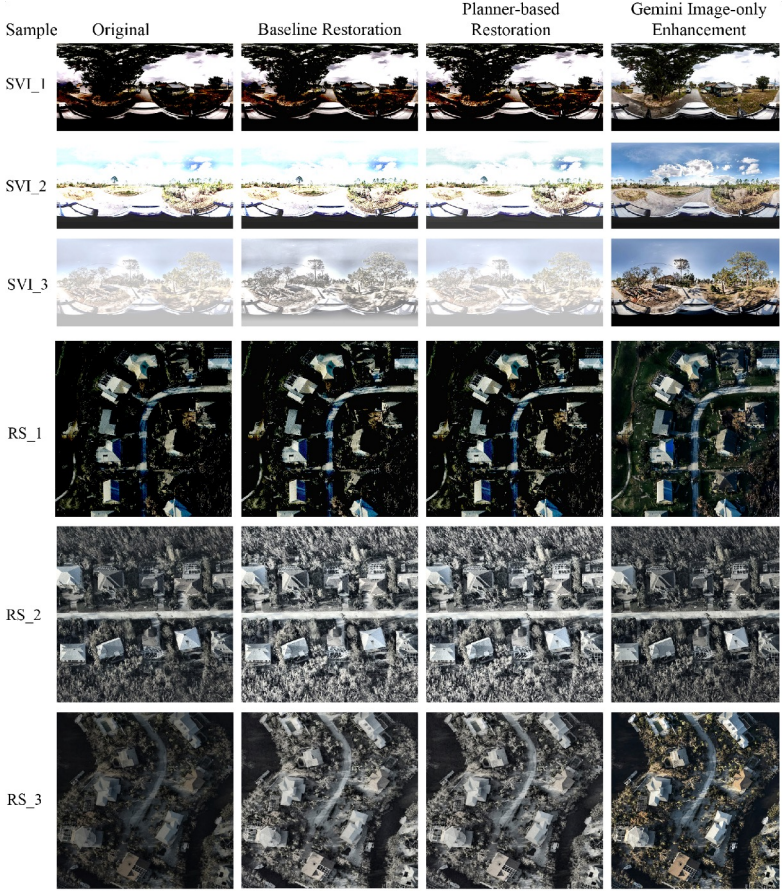

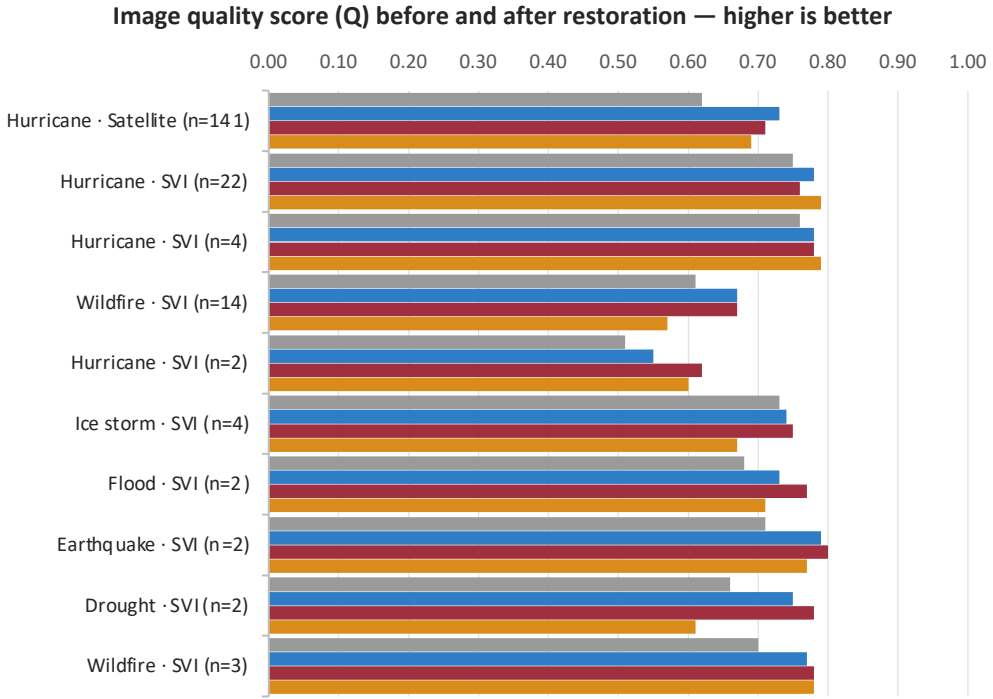

Baseline and planner-based restoration raise Q in every row (e.g., 0.51 → 0.62, 0.71 → 0.80); Gemini image-only enhancement helps hurricanes but lowers Q for wildfire, ice storm, and drought.

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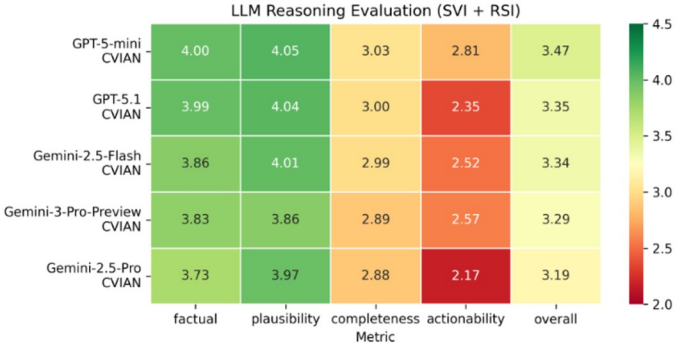
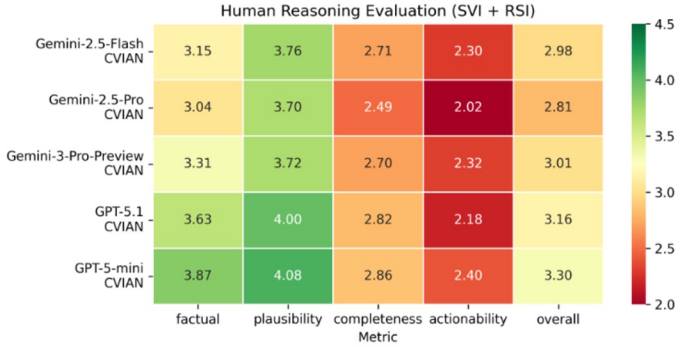
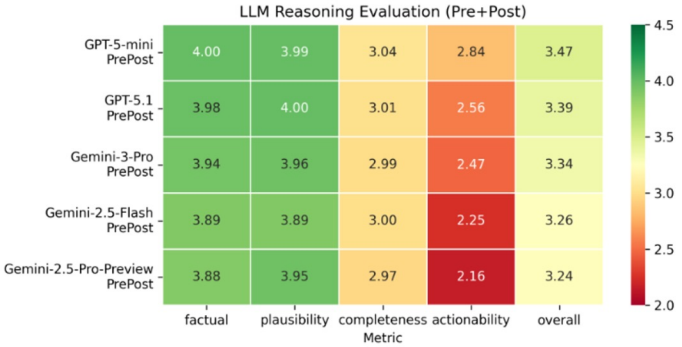
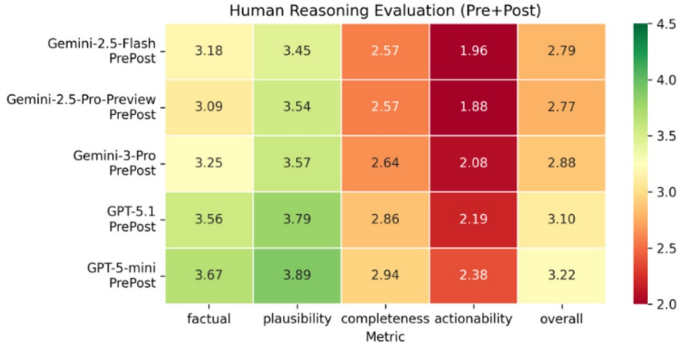
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

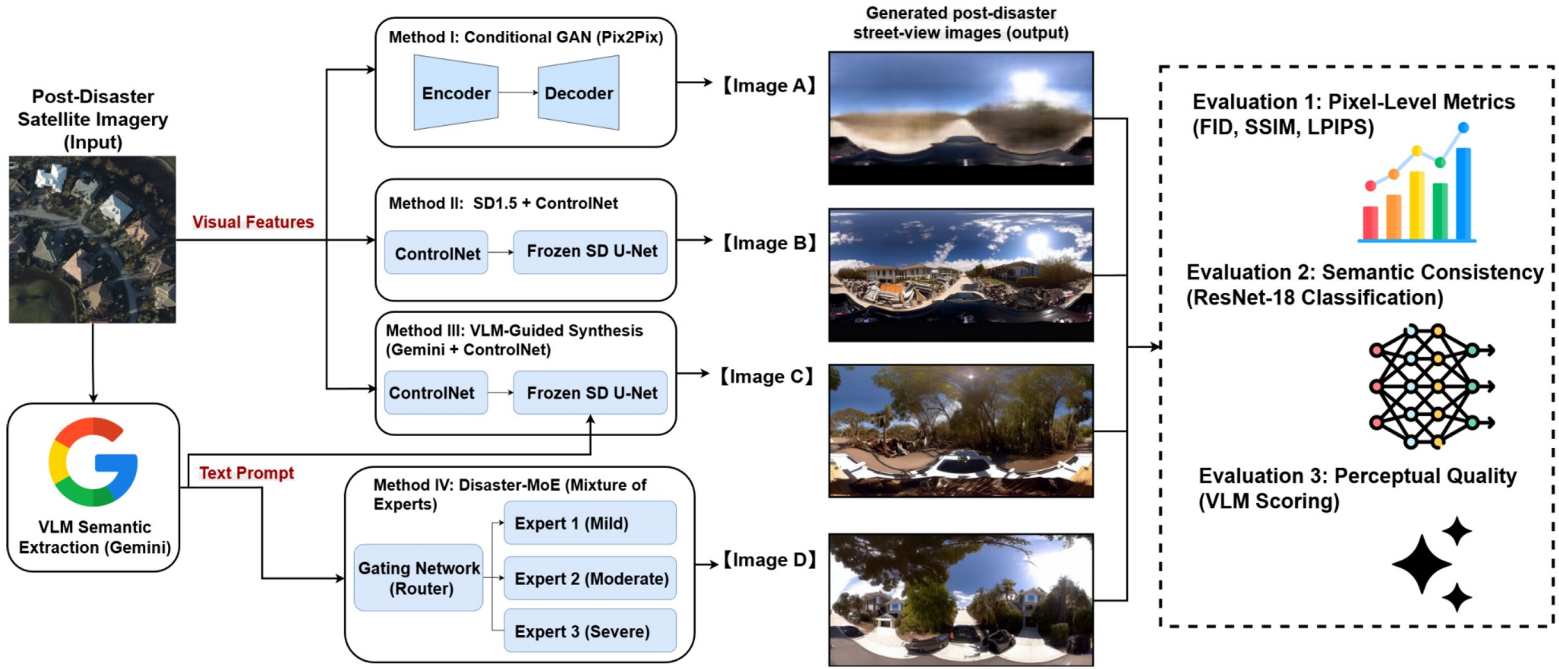






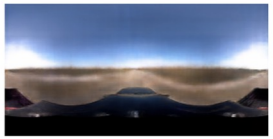
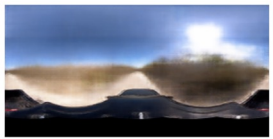


Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted, IEEE IGARSS 2026)



RQ1: Can generative vision models synthesize realistic post-disaster street views from satellite imagery?
RQ2: Do explicit disaster semantics improve synthesis fidelity?
RQ3: How can we reliably evaluate cross-view disaster synthesis beyond pixel metrics?

Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted, IEEE IGARSS 2026)

	Satellite	Pix2Pix	SD1.5+ControlNet	SD1.5+ControlNet+Gemini	MoE	Ground Truth
Mild						
Moderate						
Severe						

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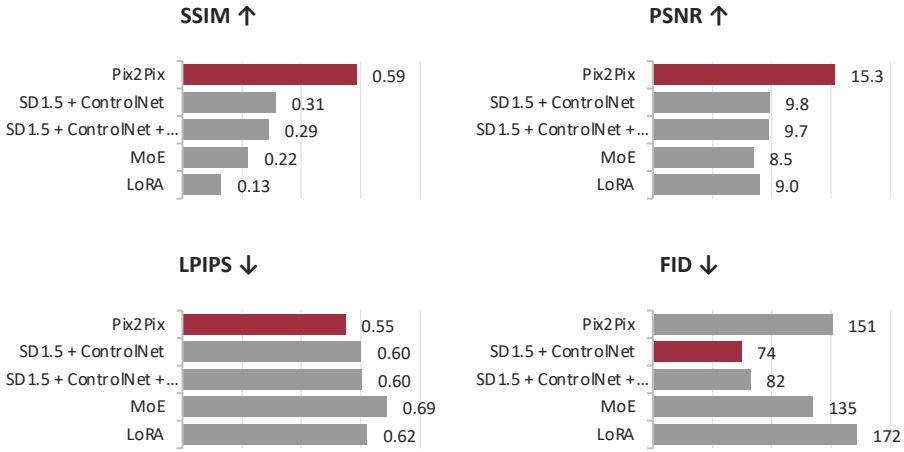
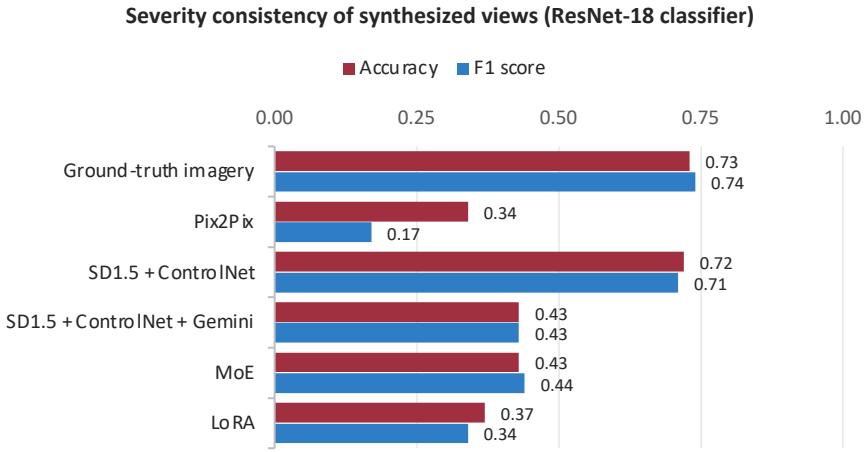
Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted, IEEE IGARSS 2026)

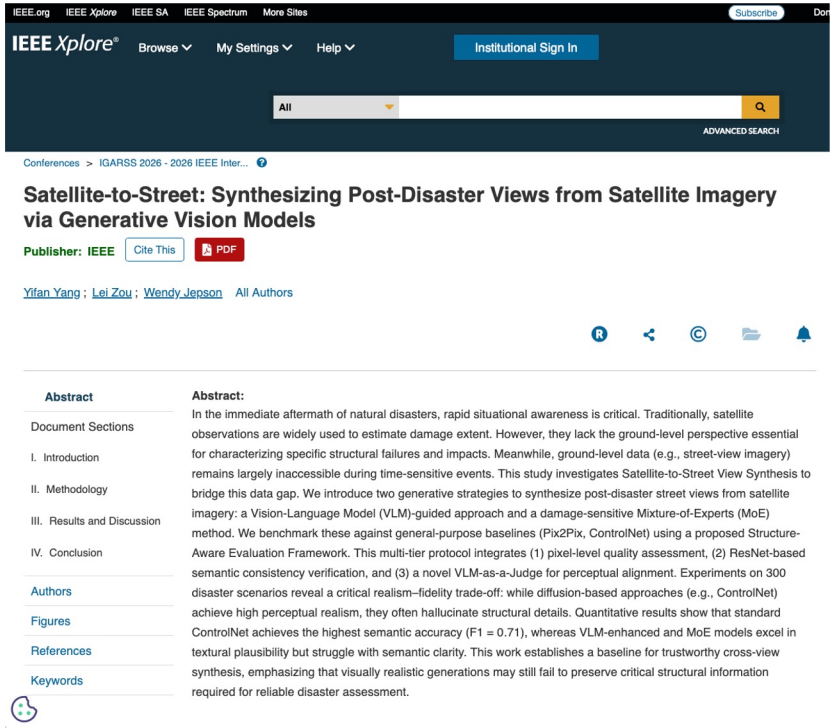




Pixel-level and perceptual image-quality metrics (best in maroon)

- Single-model approaches can't yet balance visual plausibility with strict structural alignment
- High perceptual realism ≠ high semantic accuracy — evaluation must be structure-aware
- Trustworthiness, not fidelity alone, should be the target for disaster generation

Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted, IEEE IGARSS 2026)



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Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models

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***Abstract*—In the immediate aftermath of natural disasters, rapid situational awareness is critical. Traditionally, satellite observations are widely used to estimate damage extent. However, they lack the ground-level perspective essential for characterizing specific structural failures and impacts. Meanwhile, ground-level data (e.g., street-view imagery) remains largely inaccessible during time-sensitive events. This study investigates Satellite-to-Street View Synthesis to bridge this data gap. We introduce two generative strategies to synthesize post-disaster street views from satellite imagery: a Vision-Language Model (VLM)-guided approach and a damage-sensitive Mixture-of-Experts (MoE) method. We benchmark these against general-purpose baselines (Pix2Pix, ControlNet) using a proposed Structure-Aware Evaluation Framework. This multi-tier protocol integrates (1) pixel-level quality assessment, (2) ResNet-based semantic consistency verification, and (3) a novel VLM-as-a-Judge for perceptual alignment. Experiments on 300 disaster scenarios reveal a critical realism–fidelity trade-off: while diffusion-based approaches (e.g., ControlNet) achieve high perceptual realism, they often hallucinate structural details. Quantitative results show that standard ControlNet achieves the highest semantic accuracy ($F_1 = 0.71$), whereas VLM-enhanced and MoE models excel in textural plausibility but struggle with semantic clarity. This work establishes a baseline for trustworthy cross-view synthesis, emphasizing that visually realistic generations may still fail to preserve critical structural information required for reliable disaster assessment.**

***Index Terms*—Cross-View Synthesis, Disaster Response, Generative Vision Models, Remote Sensing, Street-View Imagery.**

systematic research on generating post-disaster street views directly from satellite imagery remains limited.

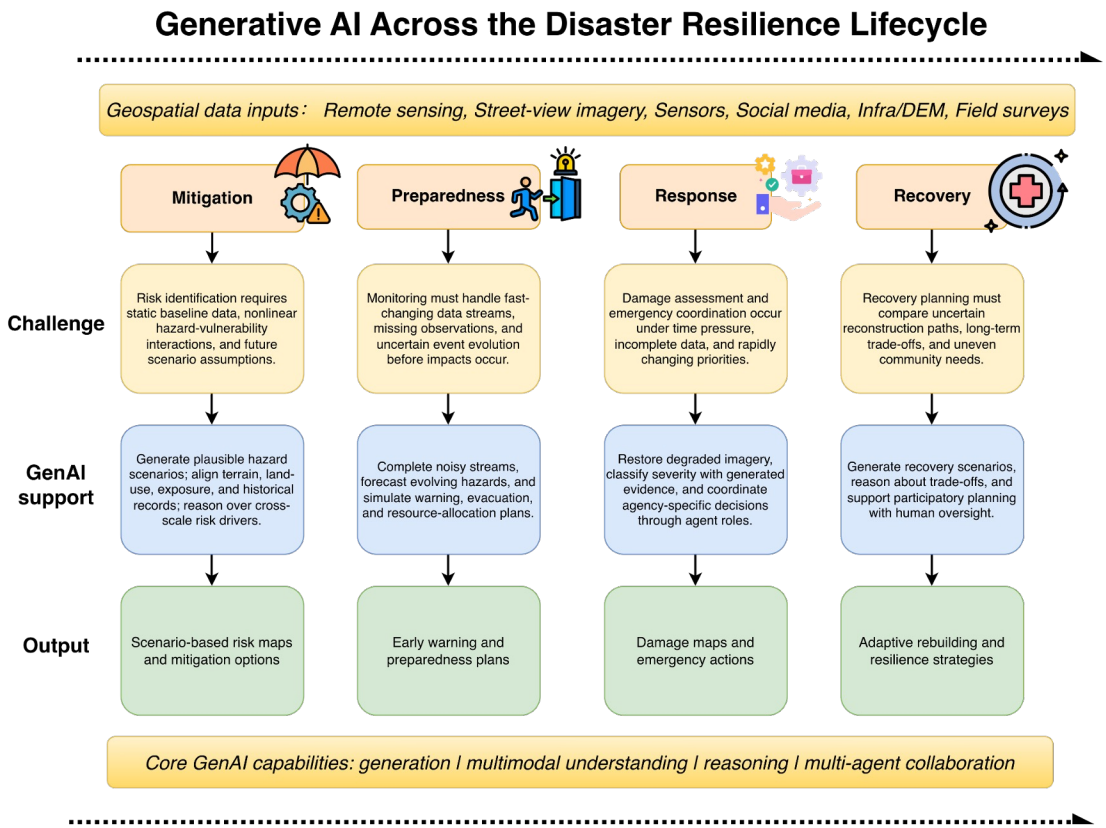
While Cross-View Image Synthesis (CVIS) has matured in urban computer vision [5], extending it to disaster scenarios introduces unique challenges. First, traditional Generative Adversarial Networks (GANs), such as Pix2Pix [6], often suffer from mode collapse in complex disaster scenes, resulting in blurred textures unsuitable for identifying wreckage details. Second, although recent diffusion-based models [7] demonstrate high fidelity, they are prone to structural hallucination in disaster contexts, referring to unintentionally "repairing" damaged buildings rather than reproducing the actual destruction in disaster scenarios. Furthermore, the inherent imbalance between damaged and undamaged samples complicates the maintenance of semantic consistency.

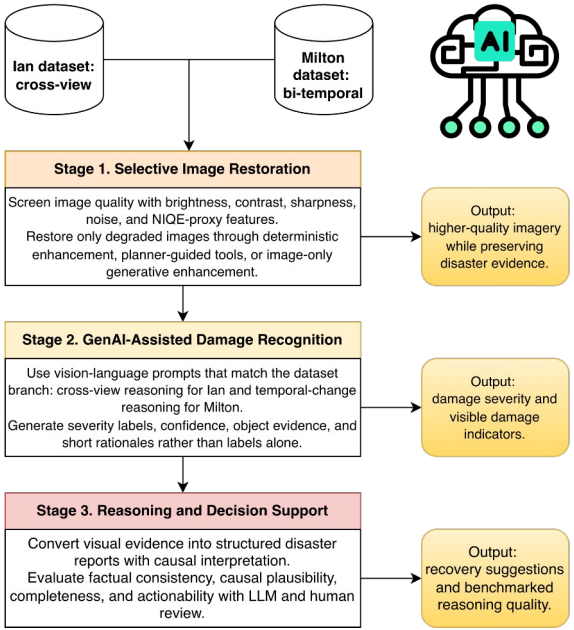
To address these challenges, this study systematically compares general-purpose baselines (Pix2Pix, ControlNet) with two newly proposed disaster-adapted strategies: a VLM-guided approach that incorporates disaster-aware semantic prompts, and a Mixture-of-Experts (MoE) framework designed to account for damage-specific visual patterns. We introduce a Structure-Aware Evaluation Framework that integrates (1) traditional Image Quality Assessment (IQA) metrics, (2) ResNet-based semantic consistency verification, and (3) VLM-driven perceptual alignment. Using this protocol, we analyze the

Y. Yang, L. Zou and W. Jepson, "Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models," IGARSS 2026 - 2026 IEEE International Geoscience and Remote Sensing Symposium, Washington DC, DC, USA, 2026, pp. 1764–1768, doi: 10.1109/IGARSS60577.2026.11730816.

Case 5: Generative AI for Disaster Resilience

(Under Review, book chapter in Geography in the Age of Generative AI, CRC Press)





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Case 8: When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment (Accepted, ACM SIGSPATIAL GeoSearch 2026)

MISSION

The proliferation of geospatial data has accelerated due to advances in Earth Observation (EO) technologies, the widespread deployment of Internet of Things (IoT) sensors, the rise of open science initiatives, and the increasing availability of cloud-native and AI-driven geospatial platforms. This data is now highly heterogeneous in format, modality, resolution, and temporal scale, and is being generated at unprecedented speed and volume. These trends introduce significant challenges—and opportunities—in how to efficiently search, integrate, and mine massive, multimodal geospatial datasets to reveal meaningful patterns and support informed decision-making.

Example topics include but are not limited to:

- Search and Retrieval using LLMs, VLMs and World Model
- Geospatial Reasoning for Effective Search and Retrieval
- Spatial Representation Learning for Search and Retrieval
- Spatial Pattern Matching
- Geospatial Knowledge Graphs for Search
- Multimedia Information Retrieval

When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment

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Abstract

Satellite and street-view imagery see different parts of a damaged building: roofs from above, facades and debris from the ground. Average accuracy hides whether fusion methods use this complementary evidence. We study the *conflict cases*, the samples on which independently trained single-view models disagree, as a diagnostic subset. On paired collections from the 2025 Eaton wildfire and Hurricanes Ian and Milton, conflicts make up 9.6–33.2% of the test data. Fusion improves conflict accuracy over either view, yet an oracle that selects the correct single view per conflict remains 0.19–0.32 above the best evaluated fusion method. A linear fusion gate over building-visibility features, calibrated confidences, and disagreement recovers part of this headroom where ground photographs are aimed at the building: on Eaton it beats end-to-end fusion by +0.072 conflict accuracy ($p < 10^{-4}$) and calibrated averaging by +0.051 ($p = 0.0001$); on the panoramic hurricane datasets it is indistinguishable from end-to-end fusion. Cropping panoramas toward the building raises the conflict gain of fusion, whereas random crops of the same size do not. The spatial density of conflicts, computed without labels, correlates with tile-level wildfire damage (Spearman $r = 0.615$, 25 tiles). Conflict cases thus show how much complementary evidence fusion uses, which view carries it under each hazard and capture setting, and where current methods fall short.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Machine learning*; • **Information systems** → *Geographic information systems.*

Collections of Geospatial Data (GeoSearch '26), November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3849732.3857333

1 Introduction

Building-level damage assessment after a disaster increasingly draws on two image collections at once. Satellite imagery covers a whole region within hours and is the basis of standardized damage mapping [2], but it mainly shows roofs. Ground-level imagery, whether crowdsourced street-view panoramas or official inspection photographs, shows the facades and debris that overhead sensors cannot fully detect [11, 13]. The views are complementary: a facade can be gutted under an unbroken roof. Methods that pair them serve geolocalization and damage mapping [5, 6, 12], multimodal arbitration [10], and agentic assessment pipelines [8, 9], and usually report that fusion improves average accuracy over either view alone.

Average accuracy says little about whether this evidence is used, because for most buildings the two views agree. The informative records are the *conflict cases*, the samples on which independently trained single-view models disagree: at least one view carries evidence the other lacks, whether damage is visible only from one side, vegetation hides the building, or the views are misaligned. Conflicts are thus a focused subset for asking how much complementary evidence fusion recovers, which view carries it under a given hazard and capture setting, and how the views can be combined.

This study examines cases in which damage assessments derived from satellite imagery conflict with those derived from street-view